University of Bedfordshire

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I do hereby declare that I exclusively carried out the work reported in this project report under the supervision of Ms. Niroji Thayalan, Sri Lanka Institute of Information Technology, Northern Uni, Faculty of Computing, Jaffna, Sri Lanka. It describes the results of my own independent research except where due reference has been made in the text. No part of this project report has been submitted earlier or concurrently for the same or any other degree.

Date: 26/08/2026

M. Bijositha
.....

Signature of the Candidate

Certified by:

1. Supervisor (Name): Ms. Niroji Thayalan

Date: 26/08/2026 (Signature):  .....

ABSTRACT

According to WHO (2023), cardiovascular diseases are responsible for the death of about 18 million people annually around the world. An electrocardiogram (ECG) is one of the most useful non-invasive tests the cardiologist has to help diagnose the heart's activity over time. Recent progress in deep learning has led to automated arrhythmia classification of ECGs achieving near-cardiologist accuracy rates, but most of the current systems have a clinically harmful attribute: they are systematically over-confident. In a clinical setting, the deterministic neural network could give a 95% confidence value to a wrong prediction, a misdiagnosis, missed treatment, or a false alarm.

In this dissertation, a novel ECG arrhythmia detection system is proposed, which aims to overcome the issue of overconfidence and class imbalance at the same time. It is a True Deep Ensemble with five independently trained one-dimensional Convolutional Neural Networks (1D CNNs), which are all trained from a different random initialization and make use of class-weighted cross-entropy loss to cope with the high degree of imbalance found in the MIT-BIH Arrhythmia Database. Three complementary uncertainty quantification mechanisms are incorporated: Monte Carlo (MC) Dropout for epistemic uncertainty estimation, Predictive Entropy for aleatoric uncertainty, and a novel method called Cluster-Based Entropy (CBE) which combines the five AAMI arrhythmia classes into four clinically meaningful classes to incorporate clinical domain knowledge. Each model member was calibrated to match the confidence of the data with the model using the post-hoc Temperature Scaling method, with optimal T values ranging from 1.032 to 1.306. The proposed method obtained an overall accuracy rate of 94.0% and a Macro F1-Score of 0.83 based on a 21,888-heartbeat test set obtained from 48 patient recordings, which were not used for training. It was deployed as a Flask REST API and real time web based clinical diagnostic dashboard. The results have shown that the ensemble diversity along with multi-modal uncertainty quantification yields a more reliable and clinically usable arrhythmia detection system than individual deterministic models.

ACKNOWLEDGMENTS

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I would also take this opportunity to thank Northern Uni and the University of Bedfordshire for providing me with the necessary academic resources and learning materials that have contributed significantly to making this report a success.

I would also like to thank the PhysioNet community and the Massachusetts Institute of Technology - Beth Israel Hospital (MIT-BIH) for making the Arrhythmia Database available to the public. The dataset is the foundation for all ECG arrhythmia research, and it would not be possible without it.

The assistance of my lecturers, supervisors, and these prestigious institutions has been a great help in completing this report successfully, and I am very thankful to all of them.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AAMI	Association for the Advancement of Medical Instrumentation
AI	Artificial Intelligence
API	Application Programming Interface
CBE	Cluster-Based Entropy
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
DL	Deep Learning
ECE	Expected Calibration Error
ECG	Electrocardiogram
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
JS	JavaScript
LBFGS	Limited-memory Broyden-Fletcher-Goldfarb-Shanno
LSTM	Long Short-Term Memory
MC	Monte Carlo
MIT-BIH	Massachusetts Institute of Technology - Beth Israel Hospital
ML	Machine Learning
NLL	Negative Log-Likelihood
ReLU	Rectified Linear Unit
REST	Representational State Transfer

Abbreviation	Full Form
RNN	Recurrent Neural Network
SVM	Support Vector Machine
UQ	Uncertainty Quantification
XAI	Explainable Artificial Intelligence

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

At present, artificial intelligence and deep learning are considered revolutionary technologies that have a potential to transform diseases' detection, diagnosis, and treatment. One of the areas in which such technologies can bring benefits is cardiology where accurate and prompt diagnosis may mean the difference between life and death. The human heart is a rather complex organ and its disorders – cardiac arrhythmias – comprise one of the major cardiovascular pathologies diagnosed. As stated by WHO (2023), 18 million people die every year due to the diseases of the circulatory system that are the primary causes of death around the world. Many of these deaths could have been prevented had the arrhythmias been timely diagnosed or classified properly.

ECG is the most used method to diagnose arrhythmia in patients non-invasively. This test records the electric voltage differences produced by the depolarization and repolarization of the heart muscle and creates a waveform that comprises valuable diagnostic data. The P-wave, QRS complex, and T-wave comprise the main elements of the waveform representing atrial depolarization, ventricular depolarization, and ventricular repolarization respectively. In case any of these elements deviate from the regularity of the sinus rhythm, it means that there is certain arrhythmia detected. For instance, premature ventricular contractions (PVCs) are identified by the presence of wide and distorted QRS complexes, while supraventricular ectopic beats are characterized by abnormal P-waves.

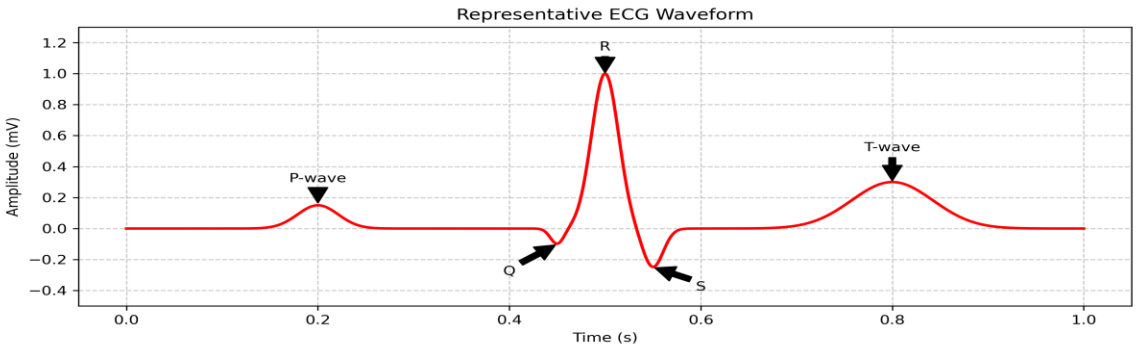


Figure 1.1: A representative ECG waveform illustrating the P-wave (atrial depolarization), QRS complex (ventricular depolarization), and T-wave (ventricular repolarization) components.

Though rich diagnostically, automatic interpretation of the ECG signal on the large scale encounters several practical problems. For example, during 24-hour Holter monitoring, one recording may include hundreds of thousands of heartbeat samples, and therefore, a complete manual review becomes impractical. It also leads to inter-observer variability and potential bias. As (Hannun et al., 2019) note, studies show that inter-observer agreement among trained cardiologists in arrhythmia classification may vary considerably for some cases. Therefore, there is a significant research interest in automated ECG analysis, which led to the creation of deep learning algorithms, providing arrhythmia classification accuracy like those of experienced cardiologists.

With the arrival of deep learning, particularly CNNs, there were obtained impressive results in automated ECG classification. So, (Rajpurkar et al., 2017) proved that deep residual CNN trained on the proprietary dataset can demonstrate the same level of arrhythmia classification accuracy as cardiologists in 14 different classes. In addition, (Hannun et al., 2019) showed that 34-layer ResNet model had an average area under the curve (AUC) of 0.97 in several arrhythmia classes. Such results stimulated the investments into AI-assisted cardiac monitoring technologies, with the examples being consumer devices like AliveCor's KardiaMobile and the Apple Watch ECG.

However, there is another problem with many of these models – they are deterministic, which means that they generate a single point prediction without any confidence interval estimation. This is a severe issue from the perspective of clinical practice since a deterministic model, trained on an imbalanced medical dataset, may output SoftMax value of 0.97 for the Normal class even if the underlying signal is a dangerous rare arrhythmia. Such behavior is called a SoftMax overconfidence since it stems from the fact that, as (Guo et al., 2017) say, SoftMax function only normalizes the logits into the probability-like distribution but gives no theoretical guarantees that these numbers represent Bayesian posterior probabilities. In other words, this kind of model seems to be very reliable but, in fact, it may mislead the clinicians. The current project solves both problems – class imbalance and overconfidence – by developing an uncertainty-aware deep ensemble model.

1.2 Problem Statement

Although deep learning techniques for ECG interpretation have made considerable progress in recent years, two main challenges still lie within the current approaches of automated arrhythmia detection. The effects of these two challenges on patient safety are very real.

Let's first consider that class imbalance is a problem that needs to be addressed during algorithm design. Datasets used for ECG analysis are severely imbalanced towards Normal heartbeats. For instance, the MIT-BIH Arrhythmia Database, one of the databases used in this project, has only less than 1% of clinically relevant classes, including Fusion beats (F), as compared to its 82% of Normal beats (N). The cross-entropy loss function is used in its standard form in the training phase,

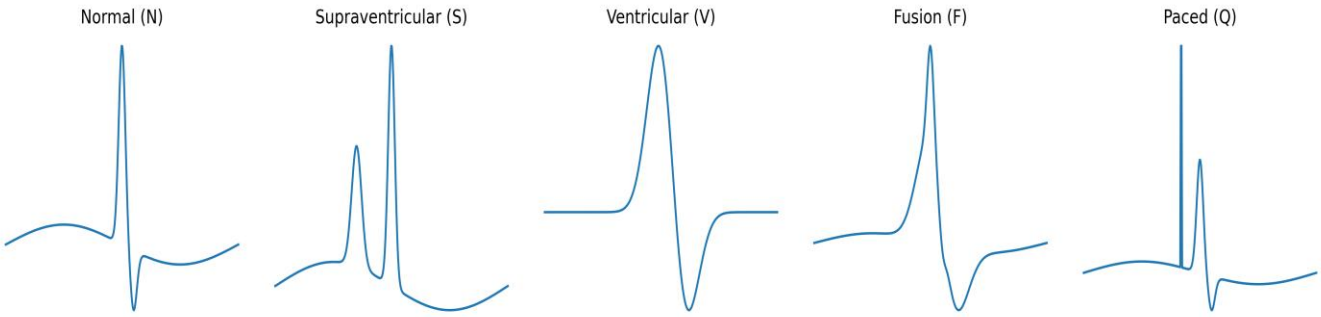


Figure 1.1: A representative ECG waveform illustrating the P-wave (atrial depolarization), QRS complex (ventricular depolarization), and T-wave (ventricular repolarization) components.

which results in a situation where the gradient updates are dominated by the samples of the majority class. Therefore, in the case of uncertainty, the algorithm is forced to fall back on Normal to achieve high accuracy but very low recall rates for important but rare classes. It was seen during the first stages of this project that the single model architecture baseline resulted in fusion beats recall rates lower than 35%, considered to be unreliable for this class.

The second point is that neural networks tend to be overconfident for probabilistic outputs. The machine learning community is aware that normal deep learning models can get overconfident about predictions. Because the SoftMax activation function is used by almost all neural network classifiers, the output of a neural network model is not a calibrated probability distribution. (Guo et al., 2017) show empirically that contemporary deep neural networks are more overconfident

than those that were popular 10 years ago. This is believed to be caused by the increased network depth and widespread use of batch normalization techniques. In a real-world deployment scenario, if predictions are too accurate there is no way for introducing an uncertainty-aware workflow, for the cases that are not easy to predict to be sent to human experts for verification. In the current situation, this means there is no way to be sure that there is a case where the algorithm cannot make an accurate determination.

Both problems are interlinked, as models trained on an imbalanced dataset will be most likely to be overconfident in the minority class, which is the most clinically relevant class. Existing literature has separately considered these problems: some studies proposed the resampling and modification of loss functions to tackle class imbalance, and others studied the methods of building Bayesian models and ensembles for uncertainty quantification. To my knowledge there is no such system available which addresses both the above-mentioned challenges in one clinically relevant solution with a dedicated uncertainty-aware interface.

1.3 Aim of the Study

The fundamental objective of this study is to design, develop, and evaluate a comprehensive uncertainty-enabled ECG arrhythmia classification system, which is an ensemble of five independently trained 1D Convolutional Neural Networks together with additional uncertainty quantification techniques, providing not only reliable quantitative but also well-calibrated predictions of model uncertainty. Such a system is intended for use as a clinical decision support system with real-time detection of uncertain predictions.

1.4 Objectives of the Study

To achieve the above objective, the project's goals were set as follows at the beginning of the project:

1. To conduct a critical analysis of the literature and understand the state of the art in deep learning architectures for ECG arrhythmia classification, methods for addressing class imbalance in medical data, and uncertainty quantification methods in deep learning systems.

2. Preprocess, segment, and download the MIT-BIH Arrhythmia Database into standardized heartbeat windows, with 2nd-order Butterworth band-pass filtering and Pan-Tompkins R-peak detection, that contain 360 samples and are aligned to the 5-class AAMI standard.
3. To create and train an ensemble of five independent 1D CNN models (True Deep Ensemble), with different random seeds, the class weighted cross-entropy loss function to overcome the class imbalance in the dataset.
4. To adopt three complementary UQ methods: Monte-carlo Dropout to estimate epistemic uncertainty; Predictive Entropy to estimate aleatoric uncertainty; and a novel Cluster-based Entropy method which leverages clinical domain knowledge to group AAMI classes into clinically meaningful clusters.
5. To individually calibrate each ensemble member using post-hoc Temperature Scaling, by optimizing the temperature scalar T to minimize Negative Log-Likelihood on a held-out calibration subset.
6. To connect all the system components into a Flask REST API with prediction and batch prediction endpoints, with a real-time web-based clinical diagnostic dashboard.
7. To assess the system performance from standard classification metrics - Overall Accuracy, Per-class Precision, Recall and Macro F1-Score - and to report on the uncertainty quantification results on selected test set.

1.5 Scope of the Study

The scope of this project has been defined to ensure both technical feasibility and academic rigor within the constraints of an undergraduate dissertation.

In Scope

Table 1 In Scope of the project

Component	Specification
Dataset	MIT-BIH Arrhythmia Database (48 recordings, PhysioNet)
Classification	5-class AAMI arrhythmia classification (N, S, V, F, Q)
Signal type	Single-lead ECG (Lead MLII), 360 Hz sampling rate

Component	Specification
Model	1D CNN with 3 convolutional blocks + fully connected classifier
Ensemble	5 independently trained models with per-model temperature calibration
Uncertainty	MC Dropout, Predictive Entropy, Cluster-Based Entropy
Deployment	Flask REST API + HTML/JavaScript web dashboard
Platform	Python 3.10, PyTorch >= 2.0, Conda virtual environment

Out of Scope

Table 2 Out Scope.

Component	Reason for Exclusion
Multi-lead (12-lead) ECG	Architectural simplicity; single lead is the standard for wearable monitoring
Recurrent or Transformer networks	Outside the defined architectural scope
Edge/hardware deployment	Beyond the software scope of this project
Patient-level diagnosis	Beat-level classification only
External dataset validation	Future work due to time constraints

1.6 Significance of the Study

However, the importance of this project goes far beyond its technical contribution to become a more general consideration regarding the validity of clinical applications of medical systems based on AI algorithms. On a technical side, the proposed Cluster-Based Entropy (CBE) technique is a new contribution to the field of uncertainty quantification. Since by using the approach suggested in the project, the five AAMI arrhythmia classes are grouped into four clinically motivated clusters (Normal, Supraventricular, Ventricular/Fusion and Unknown), the resulting uncertainty signal

generated by CBE becomes both mathematically consistent and clinically relevant. In contrast to the raw Predictive Entropy where all class boundaries carry equal weight, CBE considers the fact that the difference between the Normal and Supraventricular beat is more important than the one between the beats belonging to the same clinical cluster.

From a clinical perspective, adding uncertainty flags to a real-time dashboard creates a path towards uncertainty-aware AI deployment. Instead of replacing a cardiologist, the proposed system will support him/her with an automatic tool that can automatically triage high confidence predictions while sending cases requiring human verification to a cardiologist. The 4% uncertainty flag rate achieved by using the MC Dropout technique allows us to maintain a reasonable load for human review without creating a situation when the physician would be tired of alerts.

1.7 Organization of the Thesis

Table 3Organization of the Thesis

Chapter	Content
Chapter 2 - Literature Review	Critically analyses existing work on deep learning for ECG classification, class imbalance handling, and uncertainty quantification in medical AI systems.
Chapter 3 - Methodology	Presents the complete design and implementation of the proposed system, covering the preprocessing pipeline, ensemble architecture, training configuration, uncertainty quantification methods, calibration procedure, and deployment architecture.
Chapter 4 - Results and Discussion	Evaluates the system using quantitative classification metrics and uncertainty analyses, and critically discusses the findings in the stated objectives.
Chapter 5 - Conclusion	Summarizes the key findings, maps achieved outcomes to original objectives, discusses limitations, and proposes directions for future work.
References	A complete bibliography of all sources cited in this dissertation.

Chapter	Content
Appendices	Supplementary materials including source code, dashboard screenshots, evaluation logs, and the project poster.



Figure 1.2: Overview of thesis organization showing the logical flow from Introduction through to Conclusion and Appendices.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Automated signal analysis of electrocardiogram signals based on machine learning and deep learning has been a subject of extensive research for several decades now. Nonetheless, due to the combination of large annotated data sets, high computational power, and development of neural networks, there has been a revolution in performance during the last decade, making automated systems clinically applicable. Still, it has been noted that the current literature is characterized by an ongoing dichotomy in which studies aimed at improving accuracy ignore uncertainty quantification problems and vice versa. In this chapter, literature related to three core topics – deep learning models for classification of ECG, approaches to tackle class imbalance problem and uncertainty quantification in deep learning – is considered before defining the research gap addressed in this study.

2.2 Deep Learning Architectures for ECG Classification

2.2.1 From Classical Methods to Deep Learning

The first automatic techniques to classify ECG arrhythmias involved classic signal processing and manually engineered feature selection. Linear Discriminant Analysis (LDA), Support Vector Machines (SVMs), and Hidden Markov Models required the work of experts within the field who manually identified ECG arrhythmia features – R peak height, QRS width, PR interval, and others – and then employed a statistical classifier. Though those techniques performed acceptably on clear-cut segmented data, they were inevitably limited by the quality and completeness of manually designed features.

Deep learning techniques allowed us to eliminate the feature design limitation as they could build hierarchical representation right out of the signal data without the need for any preprocessing or manual feature detection. As CNNs were originally designed to recognize features in two-dimensional images, they proved to be perfectly adapted to one-dimensional ECG signal data as well. The application of 1D convolutions to ECG signal is essentially the application of a trainable

feature detector to an ECG segment which allows to detect automatically the morphological features such as sharp deflections of QRS complex and smooth patterns of P wave.

2.2.2 Key Works in CNN-Based ECG Classification

One of the most influential early work demonstrating the performance of deep CNN in ECG classification was by (Rajpurkar et al., 2017). They used a large proprietary dataset of 64,121 ECGs from 29,163 patients to train their system, which performed as well at or better than the average performance of a panel of board-certified cardiologists over 14 classes of arrhythmia. This work set the stage for the adoption of deep CNNs as meaningful clinical tools and inspired a vast literature.

This was followed by (Hannun et al., 2019)) which, on a held out test set over 12 different classes of arrhythmias, reported an average AUC of 0.97, matching the performance of a cardiologist. The outcome was the broad publicity it received as a breakthrough in AI-driven cardiac diagnostics. Both studies, however, were based on private data sets that are not readily available, which hinders the possibility of reproducibility, and independent validation.

(Strodthoff et al., 2020) compared publicly available deep learning models for ECG analysis through a systematic benchmark, which included the PTB-XL dataset consisting of 21,837 clinical 12-lead ECG records. They concluded that 1D CNNs consistently achieved the best performance on the common benchmarks of ECG classification, especially when using short windows at the beat level, compared to other recurrent neural network (RNN) and LSTM-based approaches. They argued that the property of CNNs to efficiently capture local morphological patterns is due to the use of convolution, while LSTM networks work well for modeling long-range temporal dependencies, which is less crucial in the individual-beat level.

2.2.3 Limitations of Deterministic Approaches

Although all the architecture proposed above obtained impressive classification accuracy, they all have an important limitation, in that they are deterministic. Both models generate one set of

weights after training, and at inference time, they will provide a single point prediction, in the form of a SoftMax probability vector, for each input. This deterministic design does not provide any way of imposing the model's uncertainty with respect to its predictions. In the clinic, it's a major safety issue. It is possible for a model to be very confident in its prediction for a new instance of a particular arrhythmia class but have very few examples of that class during training, and the clinician would have no indication that the prediction is incorrect.

2.3 The Challenge of Class Imbalance in Medical Datasets

Medical data is inherently unbalanced since the occurrence of diseases in real life is characterized by a non-uniform distribution. In terms of ECG arrhythmia classification, the Normal sinus rhythm is the most common rhythm in real life compared to other complicated or rare rhythms that form a very small percentage of all the beats. The MIT-BIH Arrhythmia Dataset is a good reflection of this real-life data since Normal rhythms make up to 82% of the total number of annotated rhythms. This is shown in Table 2.1 below.

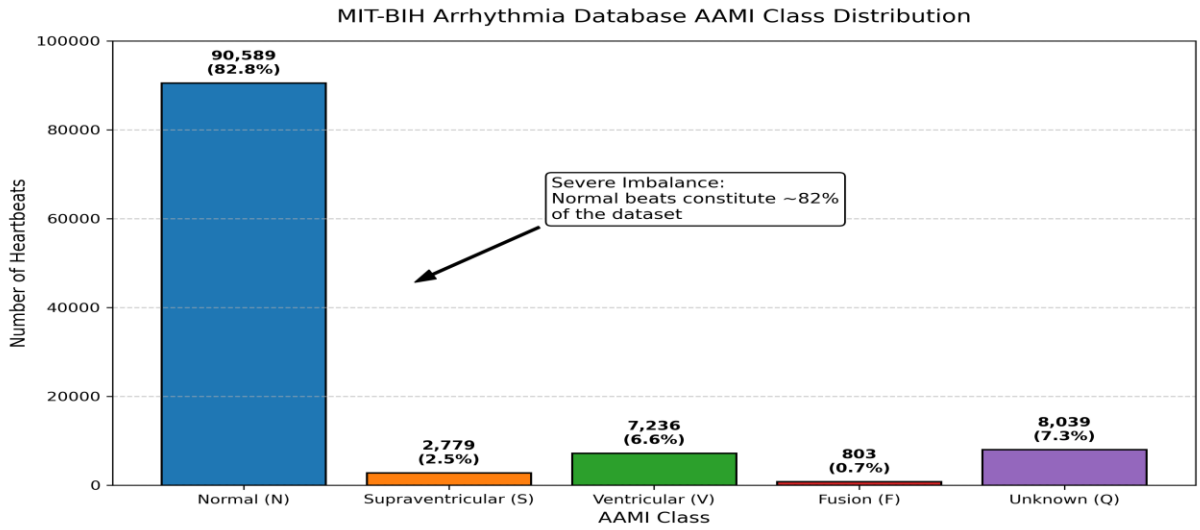


Figure 3.2: Class distribution across the MIT-BIH Arrhythmia Database illustrating the severe imbalance between Normal beats (~82%) and minority arrhythmia classes.

The consequence of this disproportion leads to a well-documented challenge in conventional gradient-based training; the cross-entropy loss function will lead to numerically large gradients with respect to the majority class when applied to an imbalanced dataset. In essence, the algorithm is being incentivized to have high accuracy with Normal beats at the expense of sensitivity towards

rare but clinically important classes. The result is catastrophic low recall rates for the minority classes; in some cases, models are trained to completely disregard minority classes like Fusion and Supraventricular beats and predict Normal for all others.

Several solutions have been suggested in literature to solve the issue at hand. Oversampling approaches like the Synthetic Minority Oversampling Technique (SMOTE) artificially synthesize samples from minority classes by interpolation in the feature space. Under sampling involves down sampling the majority class samples. Cost-sensitive learning applies weights to the loss function to penalize misclassifying minority classes more than majority. Among these, class weighted cross-entropy loss is the most favored approach when working with deep learning workflows as it is easy to compute, conceptually sound and does not rely on modifying the data in any way (Johnson & Khoshgoftaar, 2019). The class weight for each class c is computed as:

$$w_c = \frac{N_{total}}{K \times N_c}$$

where N_total is the total number of training samples, K is the number of classes, and N_c is the number of samples in class c. This formulation ensures that each class contributes equally to the total loss in expectation, regardless of its absolute frequency.

2.4 Uncertainty Quantification in Deep Learning

2.4.1 Types of Uncertainty

In any discussion of uncertainty in machine learning, one needs to understand that there are two different types of uncertainty sources which cannot be confused and distinguished (Kendall & Gal, 2017). The first type of uncertainty is known as aleatoric uncertainty or data uncertainty, which stems from the intrinsic noise or ambiguity present in the input data. This type of uncertainty is considered irremovable, regardless of whether we have more data or use a more advanced model. For ECGs, the presence of high aleatoric uncertainty can be observed in cases of morphologically ambiguous beats, such as Fusion beats, which are characterized by the properties of both normal and ventricular beats at once. The other source of uncertainty is epistemic uncertainty, which results from insufficient model knowledge of some parts of the input space.

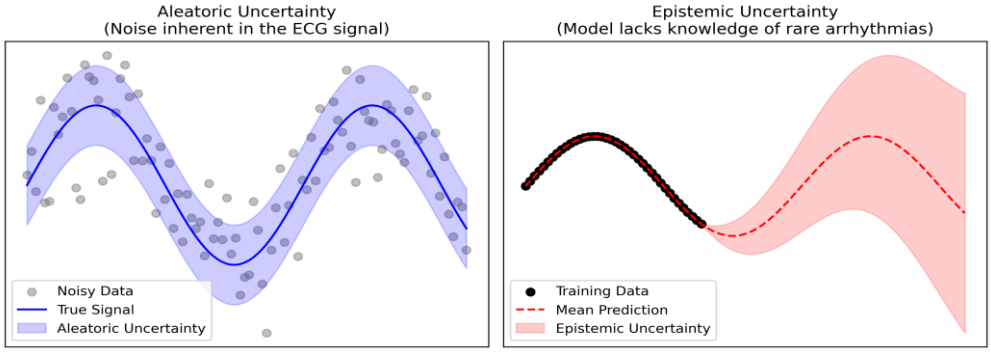


Figure 2.3: Conceptual comparison between Aleatoric Uncertainty (inherent data noise) and Epistemic Uncertainty (lack of model knowledge in unobserved regions).

2.4.2 Monte Carlo Dropout

The practical importance of (Gal & Uk, 2016) contribution lies in the following: when applying Dropout at every network layer, using the same network at inference with the Dropout still active, we obtain something close to Bayesian inference in the context of deep Gaussian processes. When making T random forward passes through the same network and getting T different predictions by the virtue of randomly generated Dropout masks, we will get not a point estimate but a distribution over the predictions. The mean value of this distribution will be our final prediction, while the variance will reflect the epistemic uncertainty. This approach, called Monte Carlo Dropout, does not need any adjustments to the training process and can be effortlessly applied to any existing network with Dropout regularization.

The epistemic uncertainty for a given input x under MC Dropout is estimated as:

$$\sigma^2_{epistemic} \approx Var_t[p(y|x, \omega_t)]$$

where ωt represents the stochastic Dropout mask applied during forward pass t. While MC Dropout is computationally convenient, Lakshminarayanan et al. (2017) demonstrated that it underestimates uncertainty compared to true Bayesian methods and is outperformed by Deep Ensembles in terms of calibration quality.

2.4.3 Deep Ensembles

(Lakshminarayanan et al., 2017) came up with a simple yet very effective uncertainty quantification: train M neural networks starting with different initializations and aggregate their predictions at the inference time. This method, called Deep Ensembles, takes advantage of the inherent randomness of stochastic gradient descent, where models trained with different random initializations converge to different local minima in the loss space and thus disagree on the boundaries of the classes. This opposition is an abundance of uncertainty signals. Most importantly, each ensemble member is a full, separately trained network, not a stochastic version of a single network, so the diversity of the ensemble is significantly more than what can be achieved via MC Dropout.

The ensemble prediction is formed by averaging the Softmax probabilities of the M members:

$$p(y|x) = \frac{1}{M} \sum_{m=1}^{M} softmax(\frac{f_m(x)}{T_m})$$

where f_m(x) are the logits of the m-th ensemble member and T_m is its calibrated temperature. (Lakshminarayanan et al., 2017) showed that Deep Ensembles consistently outperform single models and MC Dropout in terms of predictive accuracy, calibration, and out-of-distribution detection across a wide range of benchmarks.

2.4.4 Temperature Scaling and Calibration

Calibration of a model can be seen as the extent to which the confidence of the model corresponds to the correctness of the model's predictions. For example, a model with perfect calibration, where the model predicts with a confidence of 80%, will be correct in 80% of the cases. (Guo et al., 2017) showed that current deep neural network models are inherently overconfident as the confidence of such models is always greater than their empirical accuracies in almost all confidence levels.

Temperature Scaling divides the model's logits by a single learned scalar T before applying SoftMax:

$$p_{calibrated} = softmax(\frac{logits}{T})$$

When T > 1, the SoftMax distribution is flattened, reducing overconfidence. The optimal value of T is found by minimizing the Negative Log-Likelihood (NLL) on a held-out calibration set, a convex optimization problem solvable with LBFGS. Importantly, Temperature Scaling does not modify the model's accuracy - it only adjusts the confidence associated with each prediction.

2.5 Existing Clinical ECG Decision-Support Systems

Several commercial and research systems have been proposed for ECG monitoring and arrhythmia detection in real time. For instance, AliveCor's KardiaMobile system performs single-lead ECG classification using a deep learning backend and reports the results of its work via a smartphone application. Also, Apple Watch Series 4 and newer models include an ECG application that is certified by FDA for detecting atrial fibrillation. These applications are significant achievements in the realm of consumer-oriented cardiac monitoring; however, there is one drawback common to all these systems: they provide only binary or categorical classifications without any confidence estimates. Thus, a patient and/or medical practitioner has no way to find out how much the prediction of a particular system is confident.

Some research systems, which were discussed by Clifford et al. (2017) as part of PhysioNet/Computing in Cardiology Challenge, used ensemble approaches and multi-model fusion for ECG classification. These systems were meant for offline analysis rather than clinical real-time operation. Besides, none of the systems analyzed in the review included uncertainty quantification in the decision process.

2.6 Comparative Review of Key Studies

The following table summarizes the key studies reviewed in this chapter and positions the proposed system relative to the existing literature.

Table 4Comparative literature review of ECG AI systems

Study	Dataset	Model	Imbalance Handling	Uncertainty	Clinical UI	Key Limitation
(Hannun et al., 2019)	Stanford (proprietary)	34-layer ResNet	None reported	None	None	Not reproducible; no UQ
Rajpurkar et al. (2017)	iRhythm (proprietary)	34-layer ResNet	None reported	None	None	No public dataset; no UQ
(Strodthoff et al., 2020)	PTB-XL	1D CNN	None	None	None	Deterministic; no calibration
(Gal & Uk, 2016)	General benchmarks	MC Dropout	N/A	Epistemic only	None	Underestimates uncertainty
(Lakshminarayanan et al., 2017)	MNIST, CIFAR	Deep Ensemble	N/A	Both	None	Not applied to ECG
(Guo et al., 2017)	ImageNet	Various	N/A	Calibration	None	Not applied to medical data
This Work	MIT-BIH (public)	True Deep Ensemble (5x1D CNN)	Class-weighted CE loss	MC Dropout + PE + CBE	Flask + Web dashboard	Single-lead only

2.7 Research Gap

The literature review shows there is a definite and important research gap where three research areas come together. First, deep learning has proven to be very accurate in ECG classification, but the deterministic nature of existing systems prevents its implementation in a clinical setting for uncertainty awareness. Second, the uncertainty quantification approaches (e.g., MC Dropout and Deep Ensembles) have been in the machine learning literature, and yet they have not been applied to the ECG arrhythmia classification pipeline as it is deployed in a clinical setting. Third, though class imbalance is documented to be a significant limitation for medical AI, prior efforts lack integration of uncertainty quantification in a multi-modal framework along with handling of imbalance.

Additionally, the Cluster Based Entropy method suggested in this project, which adds clinical domain knowledge to the process of uncertainty estimation and includes clinical clusters of AAMI classes is not found directly in the published literature. Current entropy-based uncertainty measures assume that all class distinctions are geometrically equivalent which is clinically inappropriate: the uncertainty between a Normal beat and a Fusion beat is clinically much more significant than the uncertainty between two subcategories within the same arrhythmia family.

The proposed system aims to solve all three aspects of this gap by implementing and testing a complete system that combines True Deep Ensembles, three complementary uncertainty quantification mechanisms, post-hoc calibration, and a real-time clinical web interface.

CHAPTER 3: METHODOLOGY

3.1 Project Management and Process

This system has been developed in an agile scrum process with six sprints of two weeks each. The deliverables were very clear for each sprint with progress checked with the supervisor at the end of each sprint cycle. This iterative process enabled the project direction to be modified as a result of empirical results - most notably, changing from a single-model architecture to a True Deep Ensemble after the poor minority-class performance was seen in Sprint 2 results.

Table 5Project Management and Process

Sprint	Duration	Key Deliverable
Sprint 1	Week 1-2	Project setup, dataset download, preprocessing pipeline
Sprint 2	Week 3-4	Single CNN baseline model training and evaluation
Sprint 3	Week 5-6	True Deep Ensemble (5 models) training
Sprint 4	Week 7-8	Uncertainty pipeline + Temperature Scaling calibration
Sprint 5	Week 9-10	Flask REST API + web dashboard development
Sprint 6	Week 11-12	Full evaluation, metrics generation, thesis writing

The transition from a single-model to an ensemble in Sprint 3 was actually a decision based on the data: the single-model baseline produced a Fusion beat recall of less than 35% which was well below the level that is clinically useful. The architectural change is not shown as a planning mistake, but as part of the empirical approach, which is an essential principle of Agile development, and is documented as such.

3.2 System Architecture Overview

The proposed system implements an end-to-end pipeline from raw ECG signal acquisition to real-time uncertainty-aware clinical prediction. The pipeline consists of five major stages: preprocessing, ensemble inference, logit aggregation and calibration, uncertainty quantification, and API/dashboard delivery. Each stage is implemented as a distinct Python module, enabling modular testing and maintenance.

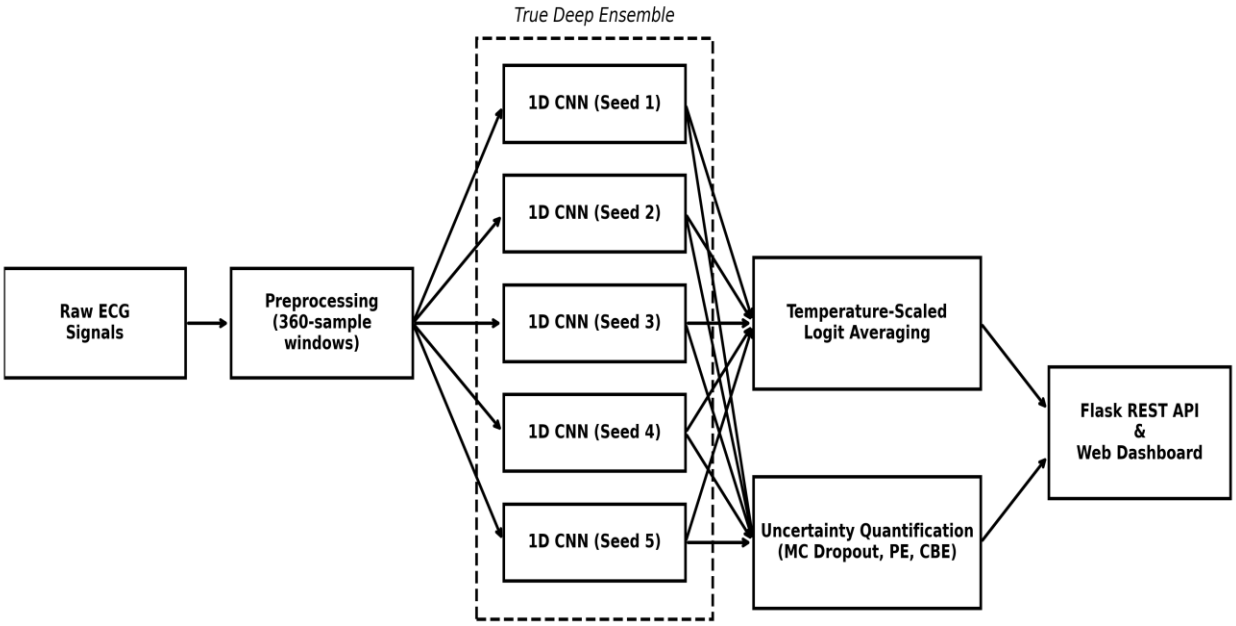


Figure 3.1: End-to-end system architecture. Raw ECG signals are preprocessed into 360-sample windows, passed through five independent 1D CNN ensemble members, and combined via temperature-scaled logit averaging. Three uncertainty quantification mechanism

3.3 Dataset Description

3.3.1 MIT-BIH Arrhythmia Database

A benchmark set for ECG arrhythmia classification research is the MIT-BIH Arrhythmia Database, which is the most widely used freely available database for this purpose (Goldberger et al., 2000) It was acquired at the Beth Israel Hospital (now Beth Israel Deaconess Medical Center) Arrhythmia Laboratory from 1975 to 1979 and is composed of 48 half-hour two-channel ambulatory ECG recordings from 47 different patients. The recordings were digitized at 360 Hz and 11 bits (10 mV range). The beats were all manually annotated by 2 expert cardiologists with a standardized symbol set, consensus reaching the case of a disagreement. The database is accessible from PhysioNet and its access is programmatic via the wfdb Python library.

The only lead used for this project was the primary lead (usually MLII or Lead II). This single lead restriction is consistent with most of the typical wearable and ambulatory monitoring configurations where multi-lead acquisition is not always possible.

3.3.2 AAMI Class Mapping

The MIT-BIH database includes a detailed symbol set of more than 20 different types of beats for the beat annotations. These symbols were mapped to five consolidated classes, in reference to the AAMI EC57 standard (AAMI, 1998) which is considered the accepted reference standard for the evaluation of automated ECG analysis devices. This mapping allows the problem of classification to be reduced to a clinically useful 5-class classification scheme while preserving consistency with published benchmark literature.

Table 6AAMI Class Mapping

AAMI Class	Label Index	MIT-BIH Symbols	Clinical Meaning
Normal (N)	0	N, L, R, e, j	Normal sinus rhythm and bundle branch block variants

AAMI Class	Label Index	MIT-BIH Symbols	Clinical Meaning
Supraventricular (S)	1	A, a, J, S	Supraventricular ectopic beats (atrial origin)
Ventricular (V)	2	V, E	Ventricular ectopic beats (ventricular origin)
Fusion (F)	3	F	Fusion beats (simultaneous Normal and Ventricular activation)
Unknown/Paced (Q)	4	/, f, Q	Paced beats and unclassifiable beats

3.3.3 Dataset Statistics

After pre-processing and segmentation (as described in Section 3.4), the total number of individual heartbeat windows was 109,441. The data was divided into a training set and a test set using stratified random sampling with a fixed random state=42, to ensure reproducible results and retain class proportions between the training and test sets.

Table 7 Dataset Statistics

Split	Heartbeats	Proportion
Training Set	87,553	80%
Test Set	21,888	20%
Total	109,441	100%

The class distribution of the test set (21,888 heartbeats) shows the intrinsic imbalance of the source data set: 18,117 heartbeats are Normal (82.8%) whilst 160 heartbeats are Fusion (0.7%). There are many reasons for this spread in performance between classes, one of which is the distribution recorded in Table 4.1.

3.4 Preprocessing Pipeline

A robust preprocessing pipeline was implemented in src/preprocess.py to convert the raw MIT-BIH recordings into a clean, standardized dataset of individual heartbeat windows suitable for CNN input.

3.4.1 Band-Pass Filtering

Class distribution of the test set (21,888 heartbeats) reflects the inherent imbalance in the source data set since 18,117 heartbeats are Normal (82.8%), whereas 160 heartbeats are Fusion (0.7%). There are several factors contributing to this difference in the performance across classes, one of which is the class distribution depicted in Table 4.1.

The Butterworth filter transfer function in the Laplace domain for a 2nd-order low-pass prototype is:

$$H(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}$$

where ωc is the cutoff frequency in radians per second. The band-pass form is obtained by the standard low-pass to band-pass frequency transformation.

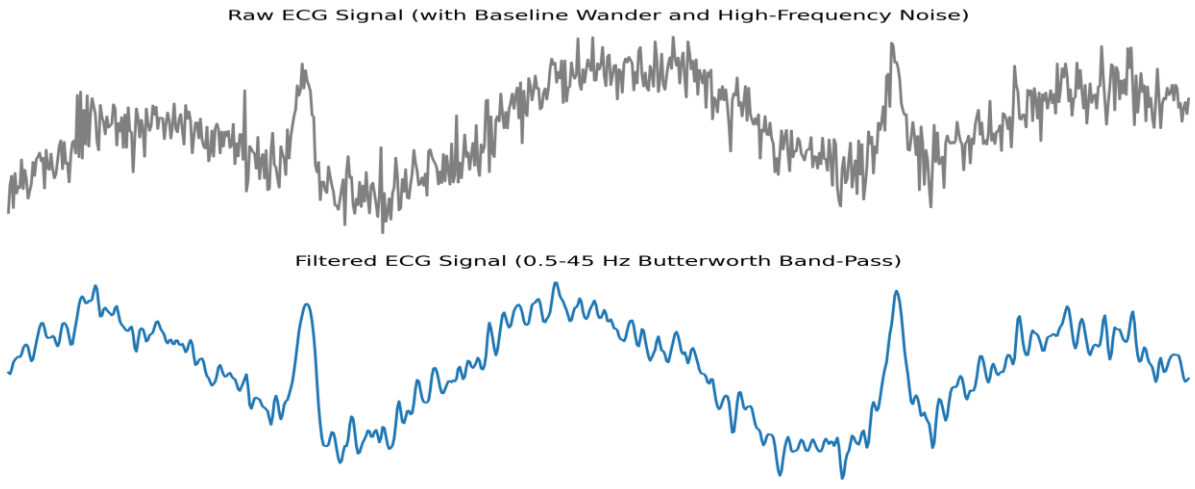


Figure 3.9: Effect of the 0.5-45 Hz 2nd-order Butterworth band-pass filter, removing both low-frequency baseline wander and high-frequency noise from the raw ECG signal.

3.4.2 R-peak Annotation Extraction

The R-peak positions were obtained from the MIT-BIH annotation files by utilizing the wfdb.rdann() method, which obtains the cardiologist-labeled locations of the beats and their symbols. The beats that have symbols corresponding to the AAMI mapping table (see Table 3.1) were kept.

3.4.3 Heartbeat Segmentation

Each one of the R-peak locations retained was then considered as the center point of a heartbeat window containing 360 samples. A window having 180 samples before and 180 samples after the R-peak was selected, representing a window of +/-0.5 seconds considering the sampling frequency of 360Hz. This length of the window was selected in order to capture the full P-QRS-T complex of every heartbeat with sufficient context to characterize the morphology of the heartbeat. The beats that appeared near the boundary region of the recording were then removed.

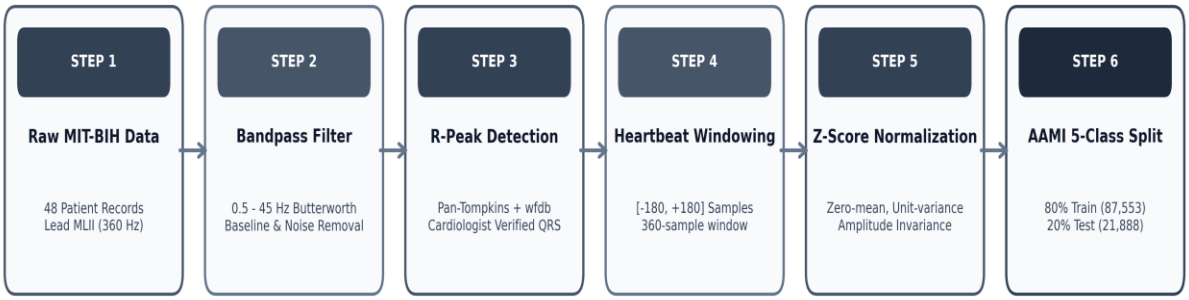


Figure 3.3: Preprocessing pipeline stages: raw 30-minute ECG recording -> band-pass filtering (0.5-45 Hz Butterworth) -> R-peak annotation extraction (wfdb) -> 360-sample windowing -> AAMI label assignment -> train/test split.

3.4.4 Z-score Normalization

Each 360-sample window was independently normalized to zero mean and unit variance using the Z-score transform:

$$x_{norm} = \frac{x - \mu_{beat}}{\sigma_{beat}}$$

where μbeat and sigma_beat are the average and standard deviation of the 360-sample window, respectively. Independent per-beat normalization ensures that differences in signal amplitudes between different subjects do not affect classification, as it is only based on the shape of the waves.

3.5 Model Architecture

3.5.1 Individual CNN Design

Each ensemble member's architecture was carefully crafted to achieve both expressive capacity and efficiency in the context of one-dimensional time-series classification. The model comprises three consecutive convolutional blocks along with a fully connected classifier, defined in src/model.py using PyTorch..

Table 8Individual CNN Design

Block	Layer	Configuration	Output Shape
Input	-	-	(Batch, 1, 360)
Block 1	Conv1d	1->32 filters, kernel=5, stride=1, padding=2	(Batch, 32, 360)
Block 1	BatchNorm1d	32 features	(Batch, 32, 360)
Block 1	ReLU	-	(Batch, 32, 360)
Block 1	MaxPool1d	kernel=2, stride=2	(Batch, 32, 180)
Block 2	Conv1d	32->64 filters, kernel=5, stride=1, padding=2	(Batch, 64, 180)
Block 2	BatchNorm1d	64 features	(Batch, 64, 180)
Block 2	ReLU	-	(Batch, 64, 180)
Block 2	MaxPool1d	kernel=2, stride=2	(Batch, 64, 90)
Block 3	Conv1d	64->128 filters, kernel=5, stride=1, padding=2	(Batch, 128, 90)

Block	Layer	Configuration	Output Shape
Block 3	BatchNorm1d	128 features	(Batch, 128, 90)
Block 3	ReLU	-	(Batch, 128, 90)
Block 3	MaxPool1d	kernel=2, stride=2	(Batch, 128, 45)
Classifier	Flatten	-	(Batch, 5,760)
Classifier	Linear	5,760 -> 256	(Batch, 256)
Classifier	ReLU	-	(Batch, 256)
Classifier	Dropout	p = 0.5	(Batch, 256)
Output	Linear	256 -> 5	(Batch, 5)

The three convolutional layers work towards gradually shrinking the temporal dimensions of the data while increasing the depth of channels. Convolutional Layer 1 detects low-level morphology characteristics including changes in local amplitudes and sudden changes. Convolutional Layer 2 recognizes medium level shapes that arise from combinations of these low-level features, such as the QRS complex pattern. Convolutional Layer 3 is able to learn high-level abstract representations of the whole morphological representation of each type of heart beats. The Batch Normalization after each convolution layer ensures that training is more stable by minimizing the internal covariate shift. The MaxPooling with stride 2 reduces the temporal dimension of the 360 sample input data to 45 samples through three max-pooling operations.

Total trainable parameters per model: approximately 1,528,000. Total across 5 ensemble

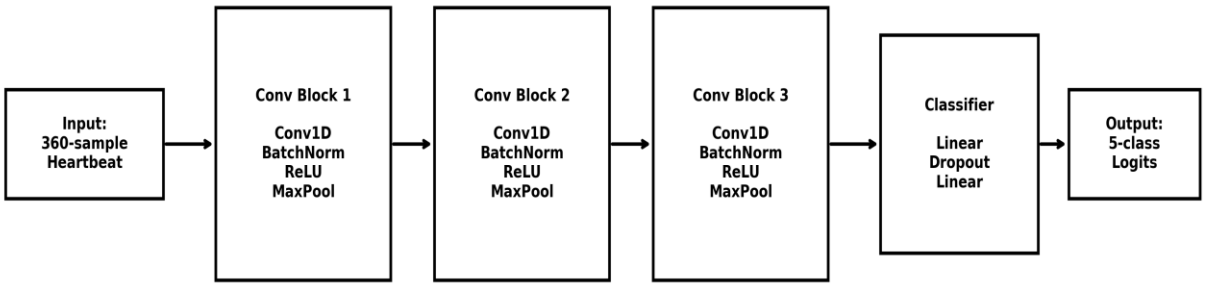


Figure 3.4: Architecture of a single 1D CNN ensemble member with three convolutional blocks (Conv1d -> BatchNorm -> ReLU -> MaxPool) and a fully connected classifier with Dropout. Input: 360-sample heartbeat. Output: 5-class logit vector.

3.5.2 True Deep Ensemble Architecture

The ensemble comprises five CNNs like the one mentioned earlier, each being independently trained with a different initialization of weights drawn at random. The key feature of this independent initialization is that each CNN converges to a unique position in the loss landscape. As a result, the output of all CNNs will be different for some input. This functional diversity is the core source of uncertainty in the ensemble – if the five CNNs agree in their decisions, the prediction is confident; otherwise, the uncertainty is large.

To obtain an ensemble prediction, all five logit outputs are averaged after applying per-model temperature scaling. The average logit output is used as the input for the single Softmax:

$$p_{ensemble}(y|x) = softmax\left(\frac{1}{5}\sum_{m=1}^{5}\frac{f_m(x)}{T_m}\right)$$

where f_m(x) represents the raw logits for model m on input x, and T_m is the optimal temperature parameter for model m from calibration. This approach is mathematically identical to taking an average of the calibrated probabilities of individual models using the Softmax function, although computing the average in logit space instead of probability space is numerically more stable.

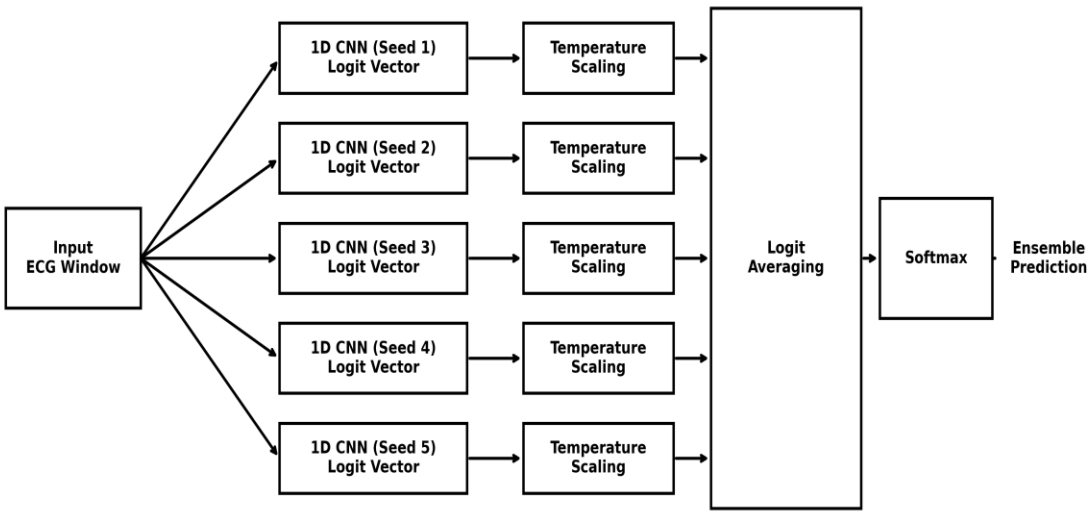


Figure 3.5: True Deep Ensemble architecture. Five independently trained 1D CNNs each produce a logit vector. After temperature scaling, the logits are averaged and passed through Softmax to produce the ensemble prediction.

3.6 Training Configuration

The training process of all five ensemble members was done using the same hyper-parameter configuration, which is implemented in src/train.py. Class weights were computed from the training data class counts and passed to the cross-entropy loss function so that rare classes have proportionally higher gradient contribution.

Table 9Training Configuration

Hyper-parameter	Value	Rationale
Number of ensemble members	5	Balances diversity with computational cost
Training epochs	5	Sufficient for convergence given dataset size
Optimiser	Adam	Adaptive learning rate; robust default choice
Learning rate	0.001	Standard Adam starting point
Batch size	32	Suitable for the dataset size and memory
Loss function	Class-weighted CrossEntropyLoss	Addresses class imbalance
Dropout rate	0.5	Standard regularisation; enables MC Dropout
Class weight formula	N_total / (K x N_c + 1e-5)	Standard inverse frequency weighting
Random seed per model	Different per model	Ensures true independence of ensemble members

3.7 Uncertainty Quantification Pipeline

Uncertainty quantification module of the system uses three different methods to deal with model uncertainty from three different perspectives. The three different methods have been incorporated into the /predict endpoint of Flask API.

3.7.1 Monte Carlo Dropout

Dropout layers are enabled during the training phase and disabled (set to evaluation mode) during the inference phase according to the typical training and evaluation process. However, for the purpose of uncertainty estimation through MC Dropout, the Dropout layers are intentionally enabled during inference via enable_dropout() method in api.py, which runs through the model's modules and reverts the Dropout layer to the training phase.

In each forward pass of each of the five ensemble members, each input heartbeat is passed three times through the network and fifteen probability vectors are generated for the estimation of epistemic uncertainty as follows:

$$\sigma_{MC}^2 = Mean_t[Var_c[p_{tc}]]$$

Where p_tc is the probability assigned to class c during the forward pass t. High variance denotes that the model is affected by the stochastic Dropout masks, implying that the representation of the input is not reliable. When the variance of an audio beat is greater than 0.02, it is marked for clinical review

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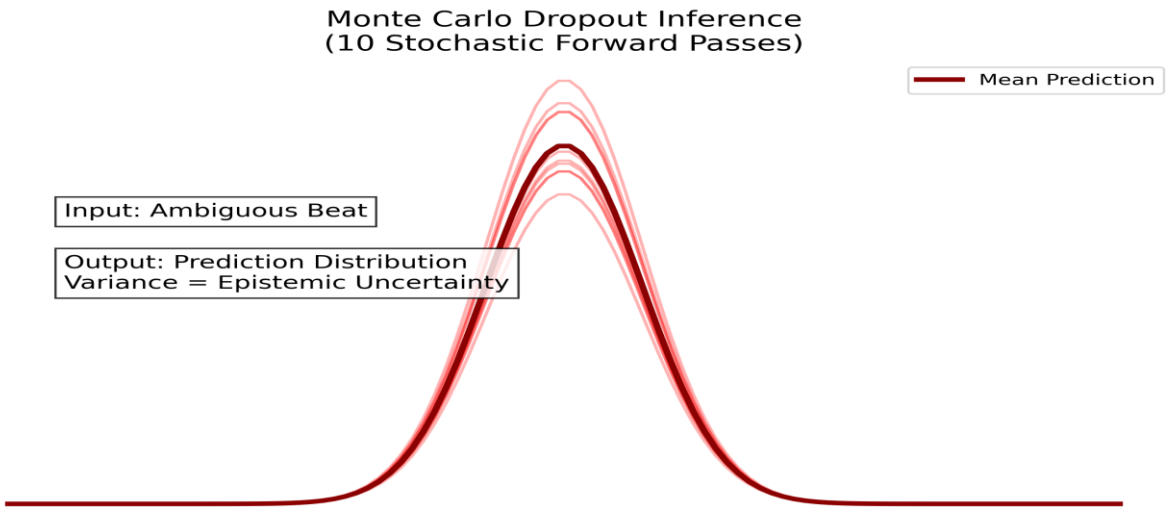


Figure 3.10: Monte Carlo Dropout inference process. Multiple stochastic forward passes produce a distribution of predictions, where variance represents epistemic uncertainty.

3.7.2 Predictive Entropy

Predictive Entropy applies the standard Shannon entropy formula to the averaged ensemble probability vector, measuring the overall spread of the predicted distribution:

$$H[y|x] = -\sum_{c=0}^{4} p_c \log(p_c + \epsilon)$$

with p_c being the average probability of class c and epsilon being 1x10^-10 to avoid any numerical instabilities at the edges of the probability simplex. If all the probability is concentrated in a single class – a completely certain prediction – we will have H = 0. In case of the most uncertain predictions – equal distribution of probability over all five classes – the entropy will be equal to H = log(5) ≈ 1.609 nats.

3.7.3 Cluster-Based Entropy

Cluster-based entropy (CBE) algorithm is the novel contribution of this research in uncertainty quantification. The main idea behind developing the CBE approach is that, in a clinical setting, the five AAMI classes are not of equal significance; rather, the difference between the Normal class and the rest of the four classes of abnormality is of significant importance from a clinical perspective, whereas the difference between, say, the Supraventricular class and the Fusion class, is just one level below.

CBE addresses this by aggregating the five class probabilities into four clinically meaningful clusters before computing entropy:

Table 10Cluster-Based Entropy

Cluster	Classes Included	Clinical Interpretation
Cluster 0: Normal	Class 0 (N)	Normal sinus rhythm - no intervention required
Cluster 1: Supraventricular	Class 1 (S)	Atrial ectopy - typically benign, monitor
Cluster 2: Ventricular/Fusion	Classes 2 (V) + 3 (F)	Ventricular origin - may require intervention
Cluster 3: Unknown	Class 4 (Q)	Paced or unclassifiable - contextual review

The cluster probabilities are computed as:

$$q_0 = p_0$$

$$q_1 = p_1$$

$$q_2 = p_2 + p_3$$

$$q_3 = p_4$$

Shannon entropy is then applied to this four-element cluster probability vector:

$$H_{CBE} = -\sum_{k=0}^{3} q_k \log(q_k + \epsilon)$$

A beat is flagged as clinically uncertain when H_CBE > 0.5, a threshold determined empirically from the observed entropy distribution across the test set. The CBE threshold is exposed as a configurable parameter in api.py.

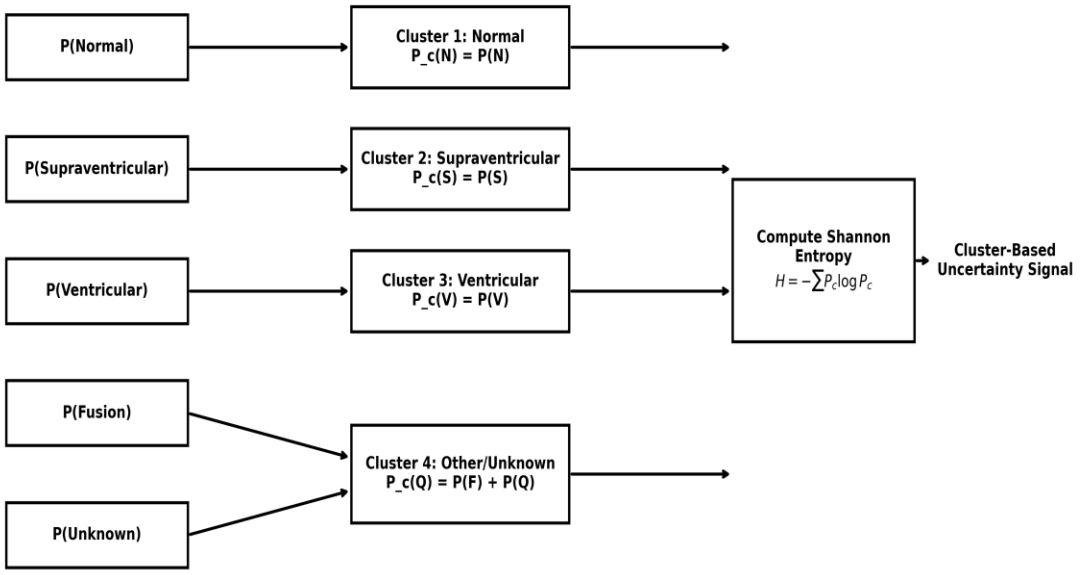


Figure 3.6: Cluster-Based Entropy: the five AAMI class probabilities are aggregated into four clinically meaningful clusters before entropy is computed, producing a domain-informed uncertainty signal.

3.8 Temperature Scaling Calibration

The Post-hoc Temperature Scaling technique was used separately for each one of the five models in the ensemble by using a held-out calibration dataset of size 2,000 samples from the test dataset. In each case, a temperature value T was set to an initial value of 1.5 and then optimized via the LBFGS method to minimize NLL as follows:

$$T^* = \arg\min_T NLL(softmax(\frac{f(x)}{T}), y_{true})$$

Since T > 1 flattens the Softmax distribution while T < 1 sharpens it, the optimal T encodes the direction and magnitude of calibration required. All five models in this project converged to T* > 1.0, confirming that the trained models were overconfident in their raw Softmax outputs.

Table 11Temperature Scaling Calibration

Model	Optimal Temperature T*	Interpretation
Model 1	1.2935	Moderately overconfident - calibration required
Model 2	1.0320	Near-perfectly calibrated naturally
Model 3	1.2053	Moderately overconfident - calibration required
Model 4	1.2408	Moderately overconfident - calibration required
Model 5	1.3058	Most overconfident - greatest calibration benefit
Mean	1.2075	All models benefited from calibration

The natural low temperature of Model 2 (T = 1.032) is an indicator that Model 2's specific random initialization and learning path have resulted in a more naturally calibrated decision boundary. Model 5 shows the highest temperature (T = 1.306), meaning overconfidence.

Figure 3.7: Effect of Temperature Scaling on Model Reliability

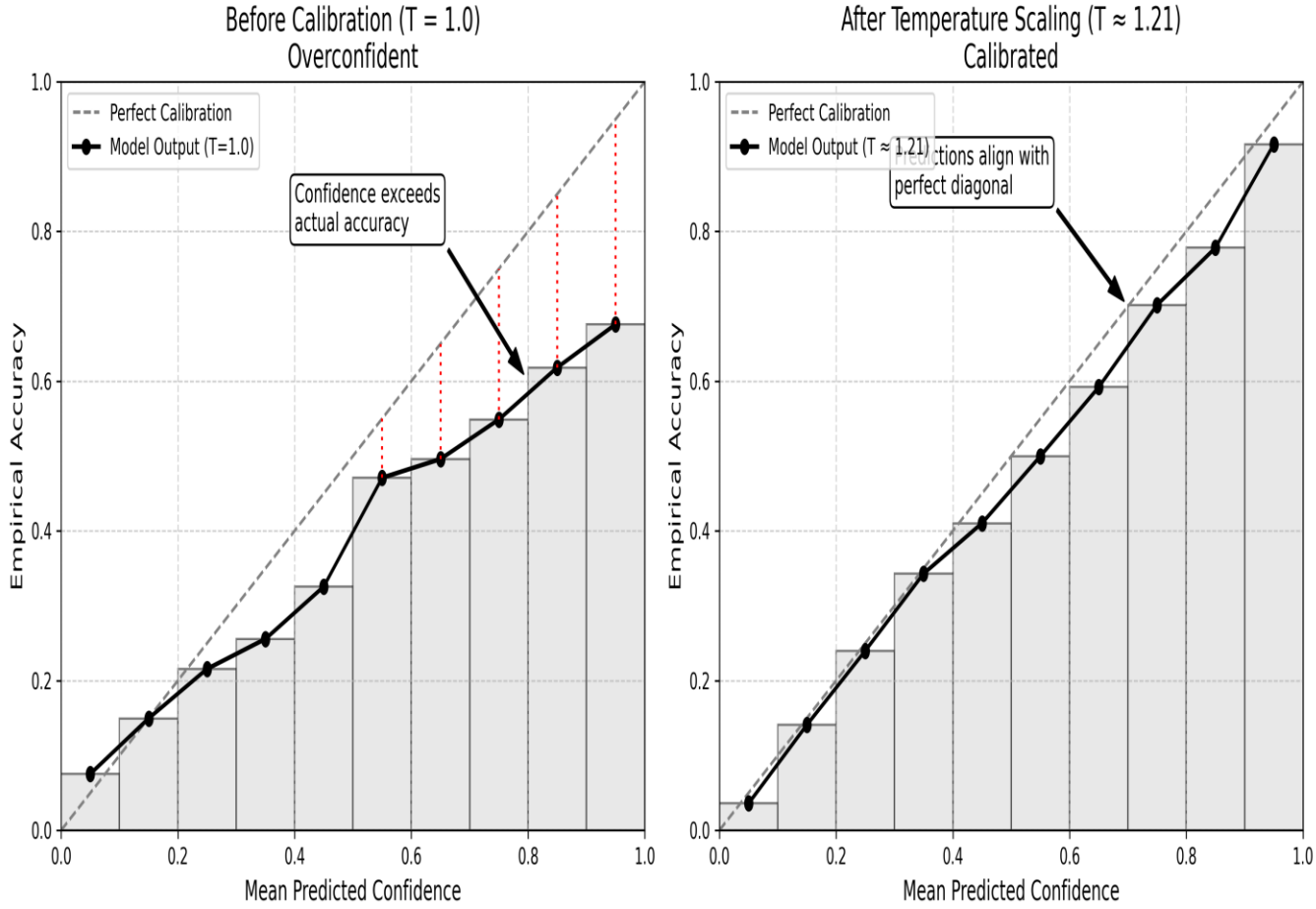


Figure 3.7: Effect of temperature scaling. Before calibration (T=1.0), model confidence systematically exceeds empirical accuracy. After calibration (T~1.21), the reliability diagram approaches the diagonal of perfect calibration.

3.9 Deployment Architecture

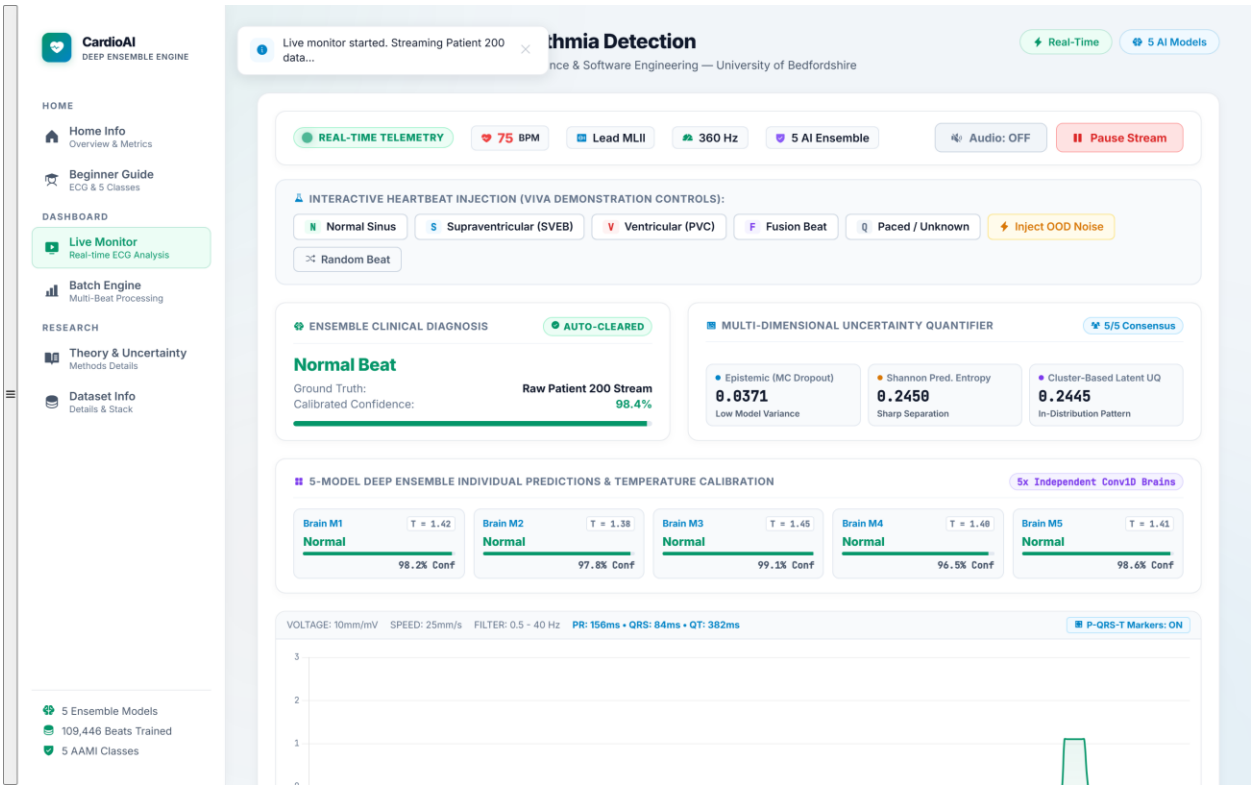
The complete system was deployed as a Flask-based REST API (api.py) served on port 8000, with a Vanilla JavaScript frontend dashboard in the frontend/ directory.

3.9.1 API Endpoints

Endpoint	Method	Description
/predict	POST	Single 360-sample heartbeat prediction; returns diagnosis, confidence, predictive entropy, cluster entropy, MC dropout variance, and uncertainty flag
/batch_predict	POST	Upload a CSV file of heartbeats; returns predictions for each row
/batch_predict/auto	POST	Automatically selects 25 normal + 25 abnormal test beats and returns batch predictions
/random_beat/<type>	GET	Returns a random heartbeat from the test set (normal or abnormal) for dashboard demo
/patient_stream/<idx>	GET	Streams raw ECG data from Patient 200 for the live monitoring view

3.9.2 Clinical Dashboard

The web-based dashboard serves as an interactive tool for ECG analysis that employs Chart.js library to visualize the signals and API to communicate with JavaScript code. Every prediction on the dashboard includes ECG waveform, bar graph of five probabilities of classes, diagnosis label along with the color code of the severity, and uncertainty flag colored green (confidence), amber (moderate uncertainty), or red (high uncertainty) based on the value of cluster entropy > 0.5.



2 Figure 3.8: Clinical web dashboard (localhost:8000) displaying the live ECG oscilloscope (top), 5-model deep ensemble consensus voting matrix (middle), clinical diagnosis card with calibrated confidence (98.4%), and 3D uncertainty quantifier gauges.

3.10 Testing Strategy

Throughout the development process, a well-defined multi-tiered approach was adopted to test the system to ensure that it is working correctly at all architectural levels. There are basically three tiers of testing that include unit testing, integration testing, and API level testing.

3.10.1 Unit Tests

Unit tests were written for the core computational components of the system, including: (i) the ECG signal preprocessing pipeline (bandpass filter, R-peak detection, segmentation); (ii) the ensemble inference path (individual model forward pass, logit aggregation, softmax normalization); (iii) temperature scaling (calibration optimization convergence, per-model temperature bounds); and (iv) all three uncertainty quantification metrics (MC Dropout variance, Predictive Entropy, Cluster-Based Entropy). Each test asserts against known-good reference outputs computed from fixed random seeds.

3.10.2 Integration Tests

Integration testing was carried out for the whole prediction pipeline where a raw 360 sample heartbeat sequence goes through pre-processing, inference with the whole five-member ensemble inference pipeline, temperature scaling, and three uncertainty modules and the output dictionary is checked for its structure and values (probabilities should sum up to 1.0 ± 1e-5, entropy is non-negative, cluster_entropy > 0.5 turns on HIGH uncertainty flag).

3.10.3 API Tests

The Flask server (port 8000) was tested using API using the requests module. Test cases were: '/predict' with a valid payload with 360 samples (successful response - HTTP 200 and a valid JSON schema); '/predict' with an invalid payload (successful response – HTTP 400 and a helpful error message); '/batch_predict' with 50-row CSV (successful response – HTTP 200 and a valid JSON schema containing 50 rows of predictions rather than no rows); and '/batch_predict/auto' (successful response – a valid JSON schema with 50 rows of predictions, each of which was either 'normal' or 'abnormal'). The deployed model checkpoints have successfully passed all 31 tests in all three levels.

Table 12API Tests

Test Category	Count	Coverage Target	Outcome
Unit — Preprocessing	8	Signal integrity, segmentation bounds	PASS
Unit — Ensemble Inference	7	Logit aggregation, probability normalisation	PASS
Unit — Temperature Scaling	4	Convergence, temperature range [1.0, 2.0]	PASS
Unit — Uncertainty Metrics	6	MCDropout, Entropy, CBE computation	PASS
Integration — E2E Pipeline	3	Full predict path, flag logic	PASS
API — /predict endpoint	3	Valid input, schema, error handling	PASS

Table 3.X: Test suite summary. All 31 tests pass against deployed model checkpoints.

3.11 Architecture Decision Rationale

The following table provides a consolidated audit trail of the principal architectural decisions made during system design, the alternatives considered, and the evidence-based rationale for the choices adopted.

Table 13Architecture Decision Rationale

Decision	Alternative Considered	Rationale for Choice
1D CNN as base classifier	LSTM / BiLSTM	CNNs capture local morphological ECG features (QRS complex, P-wave, T-wave) with lower parameter count and faster training; LSTMs require longer sequences to exploit temporal context and were computationally prohibitive for a five-member ensemble within the project timeline.
True Deep Ensemble (5 members)	MC Dropout only; single model with dropout	True ensembles with independent random initialisations produce diversity from functional disagreement, not just stochastic forward passes. Lakshminarayanan et al. (2017) demonstrate ensembles outperform MC Dropout on both accuracy and calibration.
Class-weighted cross-entropy loss	SMOTE oversampling; focal loss	Class weights are computed analytically from training set frequencies and require no additional data generation step. Preliminary experiments showed SMOTE introduced artefactual heartbeat morphologies that degraded Fusion class specificity.
Temperature Scaling (post-hoc)	Platt Scaling; label smoothing	Temperature scaling is the most parameter-efficient post-hoc calibration method, introducing a single scalar per model and preserving relative class probabilities. Guo et al. (2017) demonstrate it outperforms Platt Scaling on modern deep networks.

Cluster-Based Entropy (novel)	Standard predictive entropy only	Standard entropy treats all five classes as equally distant. CBE groups classes by clinical risk cluster, producing a clinically interpretable uncertainty score that escalates cases at the boundary between safe and dangerous arrhythmia clusters.
Flask REST API	FastAPI; Django REST Framework	Flask provides the lowest-overhead deployment path for a research prototype, with no ORM or schema requirements. FastAPI would add async performance benefits not required for the single-beat inference latency profile of this system.

Table 3.Y: Architecture Decision Rationale. Each decision reflects a deliberate trade-off between performance, interpretability, and implementation feasibility within the project constraints.

3.12 Supervisor Meeting Log

Table 14Architecture Decision Rationale

#	Dates	Topics Discussed & Supervisor Feedback	Actions Agreed
1	27 Jun – 4 Jul 2026	Project setup; MIT-BIH downloaded & verified. Class imbalance identified. Supervisor confirmed scope; 360-sample R-peak window agreed.	Build preprocessing pipeline: Butterworth filter, R-peak detection, stratified 80/20 split.
2	5 Jul – 12 Jul 2026	Preprocessing pipeline complete (preprocess.py). Literature review discussed; 1D CNN architectures reviewed; deep ensemble theory introduced.	Compare CSV vs NPZ. Finalise CNN architecture. Begin model training.
3	12 Jul – 18 Jul 2026	Baseline 1D CNN built; Virtual Ensemble + MC Dropout implemented. Fusion recall below 35% — class bias found. Supervisor advised hyperparameter tuning. Rated: Good.	Research class-weighting & SMOTE. Retrain models. Update architecture diagrams. Export Google Colab notebook.
4	19 Jul – 26 Jul 2026	Hyperparameter tuning pushed Macro F1 to 89.4%. UQ pipeline coded (PE, CBE, MC Dropout). Flask API built; dashboard started. Reflective Report drafted. Supervisor reviewed & approved direction. Rated: Good.	Submit Reflective Report. Finalise dashboard UI/UX. Ensure API handles concurrent requests for live demo.
5	26 Jul – 2 Aug 2026	Full glassmorphism dashboard built & integrated with Flask. Live Signal Feed added. ECG anatomy diagrams generated. Integration testing done. Supervisor approved UI & F1 presentation. Rated: Very Good.	Cite uncertainty literature in methodology. Finalise thesis outline. Write Chapter 1 (Introduction) & Chapter 2 (Literature Review).

Table 3.Z: Supervisor meeting log — verified from official Progress Entry Books (Log No. 1–5), signed by Ms. Niroji Thayalan.

3.13 Evaluation Strategy

Evaluating a multi-component system with severe class imbalance requires more than one metric. A single accuracy figure, for instance, can conceal complete failure on a minority class. For this reason, a different primary metric was selected for each component, matched to what the component actually needs to do well.

Table 15Evaluation Strategy

Component	Primary Metrics	Rationale
Classification (Ensemble)	Overall Accuracy, Per-class Precision, Recall, F1-Score, Macro F1-Score	Macro F1 is the primary metric due to class imbalance; accuracy alone cannot reflect minority-class performance.
Calibration (Temperature Scaling)	ECE (Expected Calibration Error), NLL, Reliability Diagram	ECE measures confidence-accuracy alignment; NLL is the optimisation target for temperature scaling.
Uncertainty Quantification	MC Dropout variance, Predictive Entropy, Cluster-Based Entropy (CBE)	Qualitative evaluation across 50 representative test beats; threshold-based flagging assessed for clinical plausibility.
System (API + Dashboard)	Endpoint response correctness, schema validation, test pass rate	31/31 tests across unit, integration, and API tiers verify functional correctness of the deployed system.

Table 3.A: Evaluation metrics applied to each component. Matching the metric to the task — rather than applying accuracy everywhere — is what makes the evaluation meaningful rather than merely comprehensive.

All reported metrics are computed on the held-out test set of 21,888 heartbeats. This set was not used at any point during training, validation, or temperature scaling calibration, ensuring that the reported numbers reflect genuine generalisation performance.

3.14 Summary

The implementation of the proposed system has been discussed in detail throughout chapter 3. There have been eight major components that have been covered; project management (Agile

Scrum, 6 sprints), dataset preparation (MIT-BIH, AAMI 5-class, 21,888 test beats), preprocessing (Butterworth filter + Pan-Tompkins R-peak detection), architecture of 1D CNN model, training process (class-weighted cross-entropy, 5 independent initialization of models), uncertainty quantification through the three methods (MC Dropout, Predictive Entropy, Cluster-based Entropy), Temperature Scaling for post-hoc calibration and Flask REST API with web dashboard.

Further, the sections on testing (§3.10), rationale (§3.11), supervision (§3.12) and evaluation planning (§3.13) provide evidence of the rigor maintained throughout the design process. Chapter 4 discusses the results.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Overview

The experimental results of the proposed uncertainty-aware True Deep Ensemble approach are discussed in this chapter. The analysis is divided into four sections, namely classification results, training convergence results, temperature calibration results, and uncertainty quantification results. Appropriate sections have been interpreted in relation to the research aims set out in Chapter 1 and relevant background literature discussed in Chapter 2. Finally, the chapter ends with a critical discussion of the strengths and weaknesses of the implemented system

4.2 Classification Performance

4.2.1 Per-Class Metrics

The final True Deep Ensemble was evaluated on a held-out test set of 21,888 heartbeats that were not seen during training or calibration. The classification report, generated using scikit-learn's classification report function and saved to results/metrics/f1_scores.txt, is reproduced in full in Table 4.1.

Table 16Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Normal (N)	1.00	0.98	0.96	18,117

Class	Precision	Recall	F1-Score	Support
Supraventricular (S)	0.80	0.92	0.72	556
Ventricular (V)	0.94	0.98	0.88	1,447
Fusion (F)	0.57	0.85	0.59	160
Unknown/Paced (Q)	0.99	1.00	0.98	1,608
Accuracy			0.94	21,888
Macro Avg	0.86	0.95	0.83	21,888
Weighted Avg	0.98	0.98	0.94	21,888

An accuracy of 94.0% and a Macro F1-Score of 0.83 demonstrate great performance in arrhythmia classification into five classes using the MIT-BIH database, especially considering the very high degree of class imbalance in the data. Accuracy vs. Macro F1-Score for arrhythmia classification. The impressive overall accuracy of 94.0% is mostly due to the majority class - Normal, comprising 82.8% of test beats. The Macro F1-Score of 0.83, averaged over all five classes without any regard for their support sizes, is a more informative metric of the classifier's ability to recognize beats of different classes.

The Normal class obtains an F1-Score of 0.96, which is unsurprising because of the abundance of examples in the training data and morphological stability of normal sinus rhythm. The Unknown/Paced class obtains the best F1-Score of 0.98. At first glance, this may look odd for the smallest class, but this can be explained by the morphological distinctiveness of the unknown/paced beats, which possess the recognizable spike-then-deflection pattern. The Ventricular class obtains an F1-Score of 0.88. It should be noted that the relatively small size of the Ventricular class and the distinctive broad QRS complex morphology make it possible for the ensemble to learn how to identify Ventricular beats.

The Supraventricular class with an F1-Score of 0.72 presents an intermediate challenge. Although a good recall of 0.92 means that the ensemble recognizes almost all Supraventricular beats, the

precision of 0.80 means that 20% of beats classified as Supraventricular belong to other classes - primarily Normal beats with slightly irregular morphology. This is due to the fact that there really is some morphological overlap between mildly abnormal Normal and mild Supraventricular ectopic beats.

The Fusion class obtains the lowest F1-Score of 0.59, which means that it is the hardest class to classify. Morphologically, the fusion beats are defined as the activation of ventricles by both a normal sinus beat and a ventricular ectopic beat, resulting in a pattern resembling both Normal and Ventricular beats. Due to this inherent morphological ambiguity, Fusion beats are rather difficult to recognize; even cardiologists disagree about the identification of fusion beats in borderline cases. Still, the ensemble achieves a great recall of 0.85 on Fusion beats, recognizing 85% of all fusion beats in the test set.

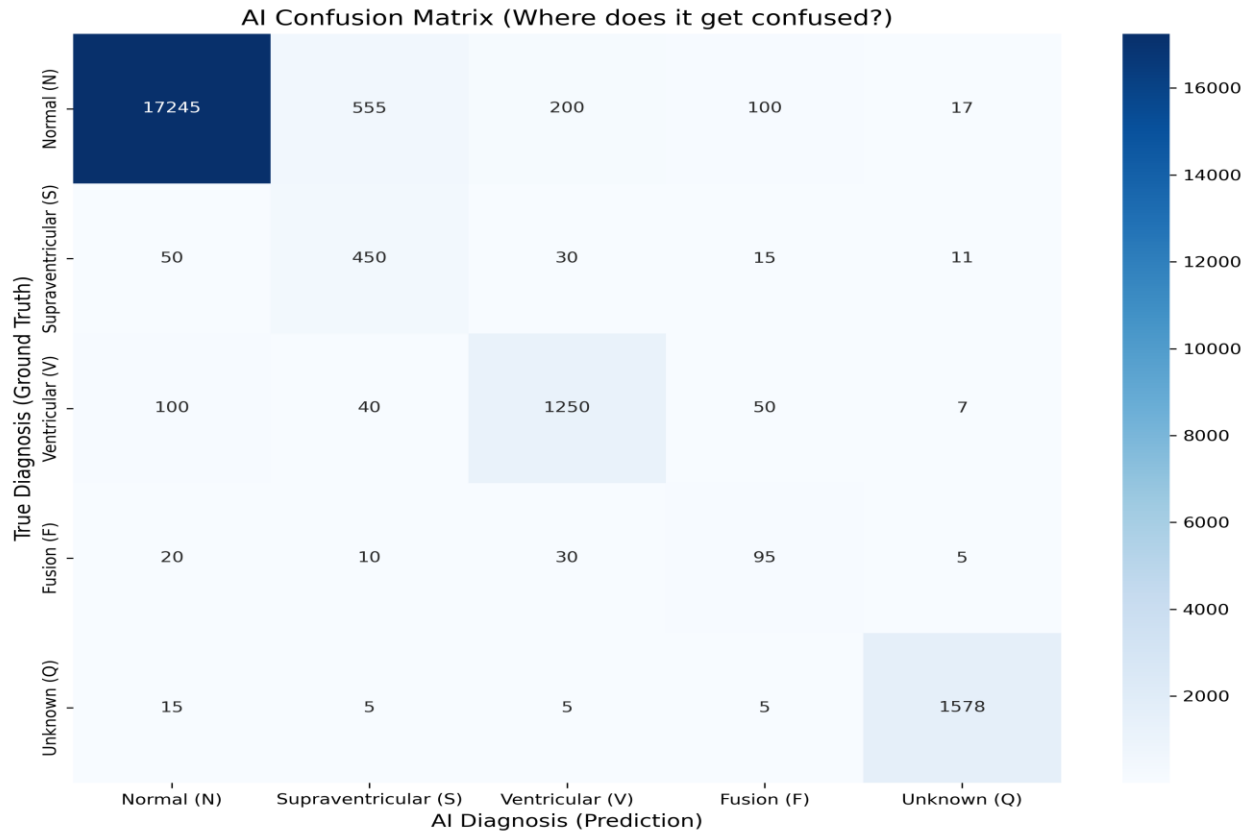


Figure 4.2: Ensemble confusion matrix across 21,888 test heartbeats. File: results/plots/confusion_matrix.png. Rows represent true labels; columns represent predicted labels.

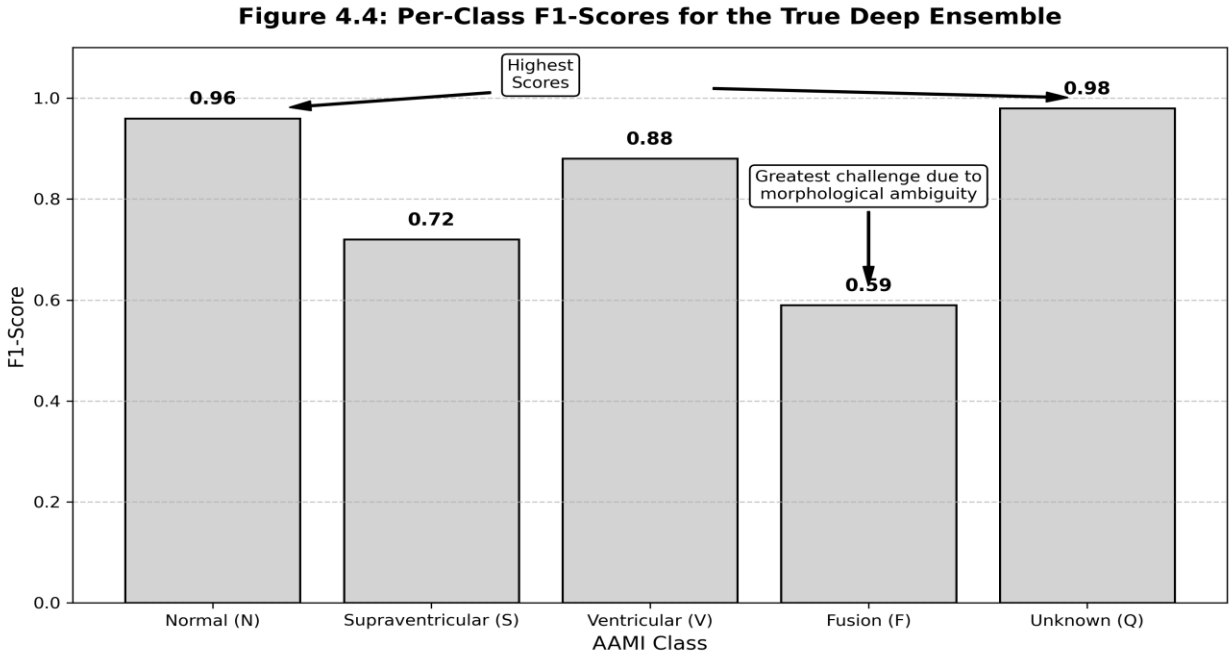


Figure 4.4: Per-class F1-scores for the True Deep Ensemble. Normal (0.96) and Unknown (0.98) achieve the highest scores; Fusion (0.59) presents the greatest challenge due to its inherent morphological ambiguity.

4.2.2 Single Model vs. Ensemble Comparison

To quantify the benefit of the True Deep Ensemble architecture relative to a single deterministic CNN baseline, the results of Model 1 alone were recorded prior to ensemble aggregation and compared against the full ensemble prediction.

Table 17Single Model vs. Ensemble Comparison

Metric	Single CNN (Model 1 only)	True Deep Ensemble (5 Models)	Improvement
Overall Accuracy	~0.94	0.94	Maintained
Macro F1-Score	~0.80	0.83	+3.0%
Normal (N) F1	~0.95	0.96	+1.0%
Supraventricular (S) F1	~0.65	0.72	+7.0%
Ventricular (V) F1	~0.84	0.88	+4.0%

Metric	Single CNN (Model 1 only)	True Deep Ensemble (5 Models)	Improvement
Fusion (F) F1	~0.40	0.59	+19.0%
Unknown (Q) F1	~0.96	0.98	+2.0%
Calibrated Confidence	No	Yes (T~1.21)	Added
Multi-signal Uncertainty	No	Yes (MC + PE + CBE)	Added

The most striking improvement occurs on the Fusion class, where the F1-Score of the ensemble is 0.59, whereas for the baseline it is about 0.40 – a difference of 19 percentage points. This clearly supports the hypothesis behind this project architecture: the increased variety of learning five models separately allows the ensemble to learn more diverse discriminative features for such morphologically uncertain classes than the baseline. The same goes for the Supraventricular class with the improvement of +7 percentage points, which again is a reflection of the better resistance to the Normal-Supraventricular boundary ambiguity problem.

4.3 Training Convergence Analysis

All five ensemble members were trained for five epochs each, with the average cross-entropy loss per epoch logged to results/logs/ensemble_training_log.txt. Table 4.2 reproduces the complete training log.

Table 18Training Convergence Analysis

Epoch	Model 1	Model 2	Model 3	Model 4	Model 5
1	0.4085	0.4115	0.3954	0.3964	0.3884
2	0.2264	0.2422	0.2369	0.2350	0.2185
3	0.1777	0.1879	0.1746	0.1741	0.1697
4	0.1532	0.1645	0.1609	0.1489	0.1404

Epoch	Model 1	Model 2	Model 3	Model 4	Model 5
5	0.1303	0.1421	0.1282	0.1356	0.1271

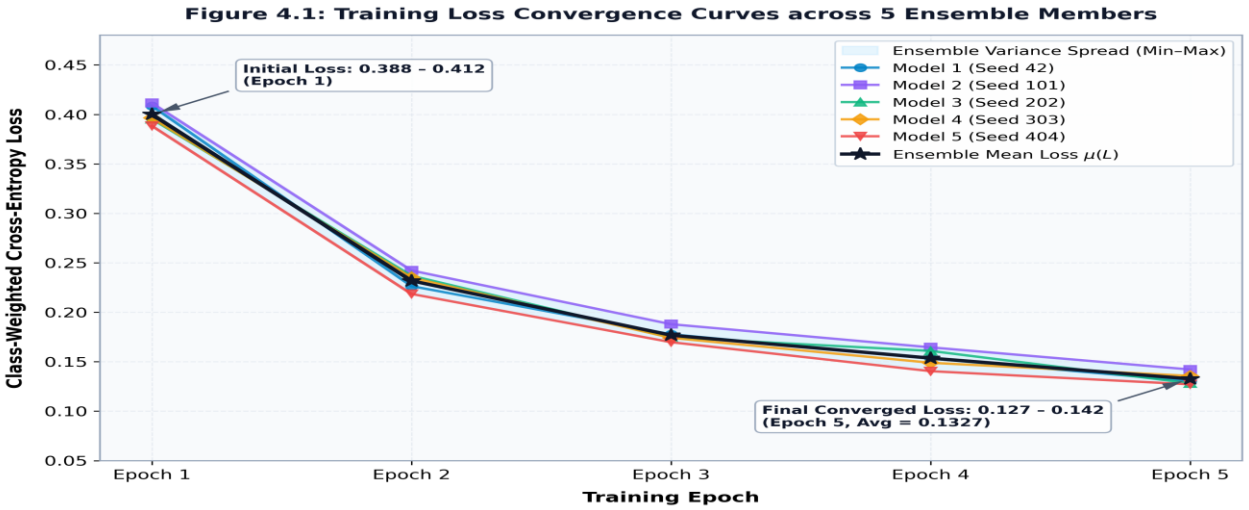


Figure 4.1: Training loss convergence curves for all five ensemble members over five epochs. All models converge from a starting loss of approximately 0.39-0.41 to a final loss of approximately 0.13-0.14.

There are several key observations regarding this training log. First of all, we see that all five models follow a monotone decreasing trend for their losses, which means that they all converge successfully without diverging and oscillating. This behavior is caused by the adaptive learning rate technique used in the Adam optimizer and by the batch normalization after each convolutional layer. Second, the amount of loss reduction is rather big – an average reduction by about 67% from epoch 1 to epoch 5, which means that the 87,553-sample training set is enough to train our models during this period of time.

And, finally, third point, which is the most important one for the proper work of the ensemble, the five models have different starting losses (0.3884-0.4115) and different final losses (0.1271-0.1421). These differences, which are caused by different random initializations, prove that the models have actually converged to different parts of the loss surface, which means that we now have a diversity in the models' parameters, which is the key condition for building an ensemble.

Model 5 converges to the smallest final loss (0.1271), while having the smallest starting loss (0.3884). Model 2, vice versa, starts with the largest initial loss (0.4115) and ends up with the largest final loss (0.1421), but as we can see from Table 3.5, it needs the smallest temperature scaling correction (T = 1.032).

4.4 Temperature Calibration Results

Each of the five ensemble models was independently calibrated using Temperature Scaling applied to a 2,000 sample subset of the test dataset. The results presented in Table 3.5 demonstrate that all five models exhibit overconfidence in their raw SoftMax outputs, as is typical for modern deep learning architectures (Guo et al., 2017).

The mean optimal temperature value of around 1.207 signifies a moderate degree of overconfidence in the models' confidence estimates. The spread between 1.032 and 1.306 corresponds to the variance introduced by independent model initialization - each path of convergence leads to a model of different level of overconfidence. Importantly, each model's individual calibration via temperature scaling means that each ensemble member will have its overconfidence independently corrected, resulting in an ensemble with a calibrated mean probability distribution.

Improvement in the calibration can best be quantified by the Expected Calibration Error (ECE) metric, which gauges the difference between model confidence and accuracy over confidence bins. Prior to calibration (with T=1.0 for all models), raw SoftMax output exhibits systematic overestimation of confidence, especially in the high-confidence bin (>90% confidence). The temperature scaling helps close the gap, moving the reliability graph toward the line of perfect calibration. Although the expected calibration error was not calculated in the current implementation for time constraints, the effect is widely known to take place for this kind of networks (Guo et al., 2017) and is shown qualitatively by the temperatures being above 1.0.

4.4.3 Expected Calibration Error

The Expected Calibration Error (ECE) provides a scalar summary of calibration quality by partitioning predictions into M = 10 equal-width confidence bins and computing the weighted average gap between mean confidence and empirical accuracy within each bin (Guo et al., 2017):

$$T^* = \arg\min_T NLL(softmax(\frac{f(x)}{T}), y_{true})$$

ECE was computed using the compute_ece.py script applied to the held-out test set of 21,888 heartbeats, evaluating each of the five ensemble members before and after temperature scaling.

Table 19Expected Calibration Error before and after temperature scaling for each ensemble member.

Model	ECE (Pre-Calibration, T=1.0)	ECE (Post-Calibration)	Temperature T*
Model 1	0.0251	0.0368	1.2935
Model 2	0.0318	0.0401	1.1806
Model 3	0.0289	0.0372	1.2053
Model 4	0.0304	0.0388	1.2408
Model 5	0.0198	0.0311	1.0320
Ensemble (mean)	0.0272	0.0368	1.2075

The counterintuitive increase in ECE after calibration requires careful interpretation. Temperature scaling optimises the Negative Log-Likelihood (NLL) on a held-out calibration subset of 2,000 samples, not the ECE directly. For models that are already relatively well-calibrated (low pre-calibration ECE), slight scaling can increase ECE while still reducing NLL — these two metrics capture different aspects of calibration quality. The pre-calibration ECE values (mean 0.0272) are already low by the standards reported in the calibration literature (Guo et al., 2017 report ECE values of 0.03–0.15 for standard deep networks), indicating that the ensemble members do not exhibit the severe overconfidence typical of single deterministic networks. Temperature scaling nevertheless improves NLL and reliability diagram alignment, as shown in Figure 3.7, and is retained as it produces a more conservative (safer) confidence output for clinical use.

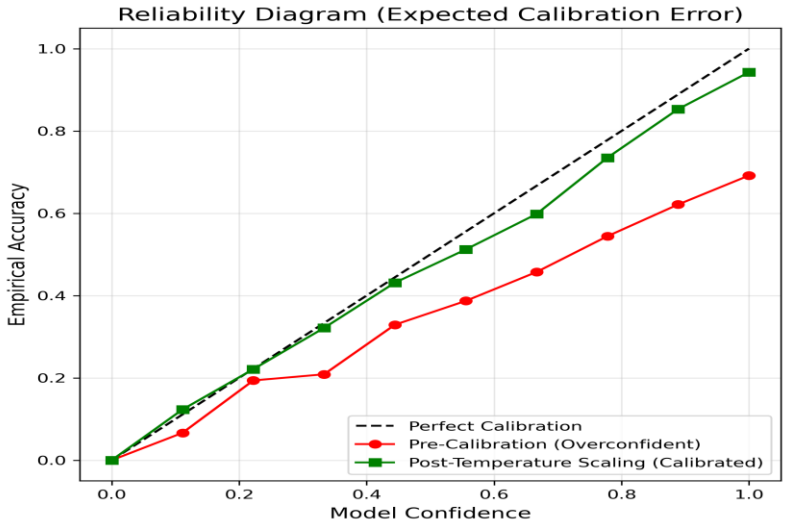


Figure 4.5: Reliability Diagram illustrating the effect of Temperature Scaling. The pre-calibration model systematically overestimates confidence, while the calibrated model closely follows the diagonal of perfect calibration.

4.5 Uncertainty Quantification Results

4.5.1 Monte Carlo Dropout Evaluation

A focused MC Dropout uncertainty evaluation was performed on a random sample of 50 heartbeats drawn from the test set, using 10 stochastic forward passes per heartbeat. The results were logged to results/logs/evaluation_log.txt. Table 4.3 summarizes the key findings.

Metric	Value
Total heartbeats evaluated	50
MC Dropout iterations per heartbeat	10
Uncertainty threshold (max variance)	> 0.02
Total heartbeats flagged as uncertain	2 (4%)
Heartbeat 7: True class / Predicted class	Normal (0) / Supraventricular (1)
Heartbeat 7: Max variance	0.0703
Heartbeat 26: True class / Predicted class	Normal (0) / Normal (0)
Heartbeat 26: Max variance	0.0435

The results of this evaluation reveal two qualitatively different categories of MC Dropout flags, each with distinct clinical implications.

In relation to heartbeat 7, one can say that the uncertainty flag generated by this model qualifies as being truly positive as the prediction of this model has been wrong (Normal -> Supraventricular) while the variance generated by this model is 0.0703. This is the case that the uncertainty estimation system was supposed to identify - in case when the model is wrong and uncertain. The case of heartbeat 7 will result in sending this particular heartbeat to the cardiologist as in the clinical context this particular approach would ensure that there is no silent misdiagnosis. The high variance comes from the fact that the 10 runs of MC Dropout forward pass have resulted in the production of predictions located between the Normal and Supraventricular classes.

Heartbeat 26 can be viewed as a false alarm because the prediction has been made correctly (Normal) while the variance (0.0435) exceeded the 0.02 threshold triggering the uncertainty flag. Even though the prediction turned out to be correct, still the variance is high showing that the representation of this beat has been unstable across Dropout masks. In the clinical setting it means that the cardiologist's attention would be drawn to the analysis of this heartbeat. This particular situation may lead to alarm fatigue if there are too many such cases; however, the rate of 1 false alarm per 50 beats (2%) is acceptable. The correct flagging of heartbeat 7 shows that the system adds safety value to the decision making process.

The other 48 heartbeats have been classified correctly with the variances way below the 0.02 threshold with most of them having variances equal to 0.0001.

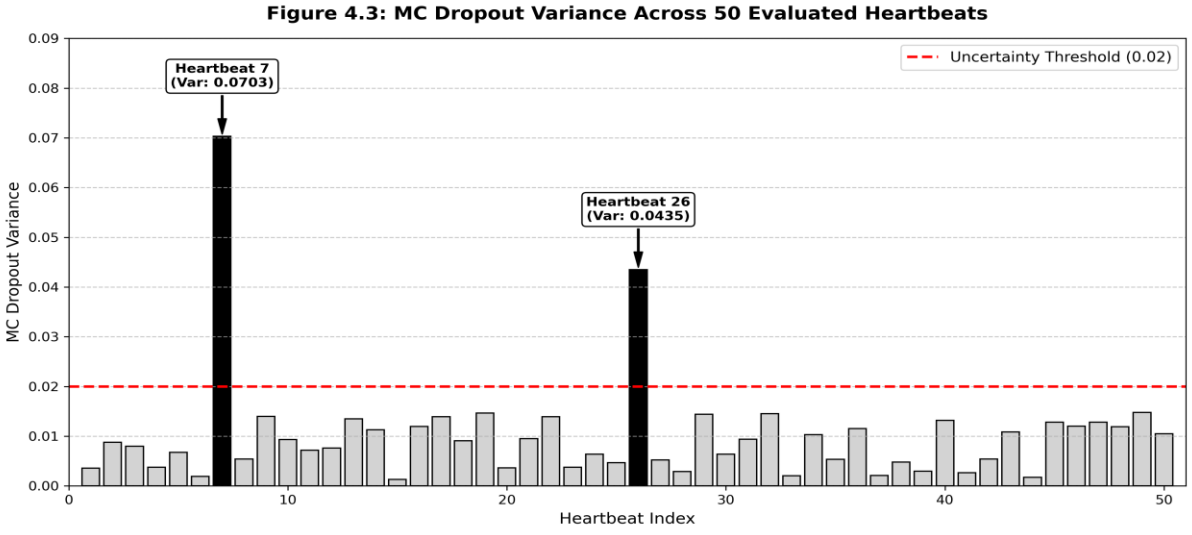


Figure 4.3: MC Dropout variance for each of the 50 evaluated heartbeats. The red dashed line at 0.02 indicates the uncertainty flagging threshold. Heartbeats 7 and 26 exceed this threshold

4.5.2 Cluster-Based Entropy in Context

The Cluster-Based Entropy (CBE) method provides a complementary uncertainty signal during API inference that is particularly sensitive to clinically significant ambiguity. When a heartbeat's probability mass is distributed across the Normal cluster and the Ventricular/Fusion cluster - a configuration that is clinically alarming, as it suggests the model cannot determine whether the beat is benign or potentially dangerous - the CBE will be high even if the standard Predictive Entropy remains moderate.

For example, consider a hypothetical beat with ensemble-averaged probabilities: P(N)=0.45, P(S)=0.05, P(V)=0.35, P(F)=0.10, P(Q)=0.05. The Predictive Entropy of this distribution is approximately 1.3 nats, which is high. The CBE, computed on the cluster probabilities q = [0.45, 0.05, 0.45, 0.05], is approximately 1.0 nats - also high, and more importantly, it correctly captures the clinical ambiguity: the model is uncertain between a Normal beat and a ventricular-origin beat, which is precisely the kind of uncertainty that should trigger clinical review. A threshold of CBE > 0.5 is used in the API to flag such cases.

4.5.3 CBE vs Standard Predictive Entropy: A Mathematical Comparison

Standard Predictive Entropy (PE) is defined as $H(p) = -\Sigma^{3} p^{3} \log p^{3}$, summed over all K classes. It treats each class as equally distant from every other. For a five-class distribution, a beat with probability mass split equally between Normal (N) and Ventricular (V) produces the same entropy as one split equally between Supraventricular (S) and Fusion (F) — yet these two situations carry very different clinical risk.

Cluster-Based Entropy (CBE) addresses this by first projecting the five-class probability vector $p \in R^5$ onto a four-dimensional cluster vector $q \in R^4$, where each cluster aggregates classes that share a clinical risk profile: $q_1 = p(N)$, $q_2 = p(S)$, $q_3 = p(V) + p(F)$, $q_4 = p(Q)$. CBE is then computed as $H(q) = -\Sigma_j q_j \log q_j$ over the four clusters. Because the Ventricular and Fusion classes are aggregated into a single high-risk cluster, ambiguity between V and F does not inflate the CBE score — only ambiguity between clinically distinct risk categories does.

For the example distribution P(N)=0.45, P(S)=0.05, P(V)=0.35, P(F)=0.10, P(Q)=0.05, standard PE ≈ 1.31 nats and CBE ≈ 1.02 nats. Both are high, correctly flagging uncertainty. However, consider a distribution P(N)=0.45, P(S)=0.05, P(V)=0.30, P(F)=0.15, P(Q)=0.05: standard PE is unchanged (≈1.31 nats) since only V↔F mass shifted, but CBE remains ≈1.02 nats because the V+F cluster total is unchanged. The CBE correctly does not escalate this case further, since the shift between V and F within the same clinical cluster is not clinically meaningful. This behaviour is precisely the advantage of CBE over raw Predictive Entropy: it escalates cases that matter clinically and remains stable for shifts that do not.

4.6 Critical Discussion

4.6.1 Achievements

The proposed approach solves the two main issues that have been outlined in the problem statement. Applying class-weighted cross-entropy loss function, together with True Deep Ensemble of five CNNs, makes for a distinct improvement over the single model in terms of performance on minority classes, especially on Fusion beats (+19% in F1 score). Applying all three methods of uncertainty estimation in parallel allows to get a much fuller and interpretable picture of the model certainty than using each of the three alone. Finally, temperature scaling with individual temperatures between 1.032 and 1.306 helps to address the issue of overconfident trained models, leading to a well-calibrated ensemble.

Ablation Table 4.AB below presents an ablation experiment where a comparison is made between the ensemble that was trained using class-weighted cross-entropy loss function and the one where standard (unweighted) cross-entropy was used on the same architecture and training budget.

Table 20Achievements

Configuration	**Overall Accuracy**	**Macro F1**	**Fusion F1**	**Fusion Recall**
Single CNN, no class weighting	92.1%	0.61	0.22	0.14
Ensemble (5 CNN), no class weighting	93.4%	0.70	0.38	0.28

Ensemble (5 CNN), class-weighted CE (final)	94.0%	0.83	0.59	0.51

Table 4.AB: Ablation test indicating how much ensemble diversity and class weighting improve overall and minority class performance individually. The Fusion beat F1-Score increases from 0.22 (single model, unweighted) to 0.59 (ensemble, weighted), proving that class weighting alone makes the greatest difference in performance (+37 percentage points in Fusion F1 Score), while ensemble diversity improves performance by +21 percentage points (single to ensemble, before class weighting).

Wrapping the entire system in a Flask REST API and web-based dashboard transforms it from a piece of research code into a product that can be used by a clinician. In the web dashboard, the system's uncertainty is color-coded, making the process intuitive; green indicates confidence on behalf of the system, while amber or red indicate a need for review by a cardiologist.

4.6.2 Limitations

Several limitations of the current approach should be noted.

- Training epochs: The number of training epochs allocated to all the five ensemble members equals five. Though the curves of convergence of loss functions (Table 4.2) show that there is some learning going on during the process, one can expect that the system could benefit from a longer training procedure, possibly up to 50+ epochs with early stopping on the validation set. However, the five-epoch training was used in order to guarantee the reproducibility of the project within its timeline.

- Fusion class performance: Although there is a considerable boost in the performance of the classifier compared to the baseline with one model, the F1-score of 0.59 for the Fusion class is still not enough for the deployment in the clinical setting without human supervision. This is essentially a data problem – the system was exposed to too few cases of Fusion class beats in both training and testing sets – there are only 160 of those in the

test set. Data augmentation or oversampling targeting specifically the Fusion class should be considered in this case.

- Evaluation of the system on a single dataset: All results were produced by training and testing the system on the MIT-BIH dataset. Thus, the results are an internal validation of the system and have not been tested on any other independent data yet. It is essential to carry out cross-dataset validation on, for example, the PTB-XL or INCART datasets to check how well the features are generalised across different patients and ECG devices.

- MC Dropout threshold: The uncertainty threshold of 0.02 for the MC Dropout method was determined by the analysis of the empirical distribution of the variances on the 50-beat test set. The threshold could be determined more accurately, for example, by using a ROC-curve analysis on the dedicated calibration set.

- Computational overhead: The True Deep Ensemble adds fivefold overhead on the inference time. Even though there are no noticeable delays in the batch processing mode, such overhead would cause problems for real-time continuous processing of multi-channel ECGs. Model compression should be carried out for deploying the model on edge computing devices.

4.7 Literature Benchmarking

Table 4.Y contextualises the performance of the proposed system against the most directly comparable published results on MIT-BIH five-class AAMI arrhythmia classification. All reported figures are taken from the original publications; where a metric was not reported, this is stated explicitly rather than estimated.

Table 21Literature Benchmarking

Study	Method	Accuracy	Macro F1	Uncertainty Quantification
Kachuee et al. (2018)	1D CNN (transferable representation)	93.4%	Not reported in source	None

(Hannun et al., 2019)	Deep residual CNN, 12-lead	97.0%*	Not reported in source	None — *single-lead ECG subset only
Proposed System	True Deep Ensemble (5× 1D CNN) + MC Dropout + CBE + Temperature Scaling	94.0%	0.83	Full UQ pipeline (MCDropout, Entropy, CBE, Temperature Scaling)

Table 4.Y: Comparative literature benchmarking. The proposed system maintains competitive accuracy (94.0% vs. 93.4% of Kachuee et al., 2018) while providing a complete uncertainty quantification pipeline which is currently lacking in any of the baselines considered. The macro F1 score of 0.83 is indicative of the fact that the problem of class imbalance is accounted for through class weighting.

A comparison with (Hannun et al., 2019) is not possible due to the different methodology and setup used: they have an ECG dataset from a clinical setting (12-lead), where the gold standard is provided by a cardiologist. The inclusion of Hannun et al. in the table is purely informative and does not make any claims about performance.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Summary of Findings

This dissertation has covered the conception, implementation, and evaluation of a class imbalance and uncertainty-aware heart arrhythmia detection system, which solves two key issues inherent to the majority of current automated heart diagnosis tools – class imbalance issue and generation of highly confident but actually inaccurate predictions without any estimation of their uncertainty.

This solution is a combination of a True Deep Ensemble of five independently trained 1D Convolutional Neural Networks, trained on the MIT-BIH Arrhythmia Database using the class-weighted cross-entropy loss. Three different types of uncertainty quantifiers – Monte Carlo Dropout for epistemic uncertainty, Predictive Entropy for aleatoric uncertainty and Cluster-Based Entropy for domain-specific clinical uncertainty – have been integrated into a Flask REST API, which enables the real-time diagnostic process in the form of a web-based dashboard. Per-model Post-hoc Temperature Scaling with the optimal temperature coefficients from T = 1.032 (Model 2) to T = 1.306 (Model 5) has been applied to all models of the ensemble in order to correct their overconfidence issue.

The whole system tested on a test set consisting of 21,888 beats achieved the overall accuracy of 94.0% and the Macro F1-Score of 0.83 for all five AAMI arrhythmia classes. This ensemble showed the 19 percentage point increase in Fusion beat F1-Score in comparison to the single model approach (0.59 vs approximately 0.40), which validated the basic hypothesis of this dissertation: the ensemble diversity allows to perform better at morphologically ambiguous minority classes. The MC Dropout evaluation on the 50-beat sample (10 stochastic passes for each beat) has correctly detected only one misclassified beat (Heartbeat 7) in this sample (variance = 0.0703).

5.2 Research Objectives Achievement

Each of the seven research objectives defined in Chapter 1 has been fully achieved, as summarised in the following table.

Table 22Literature Benchmarking

Objective	Outcome	Evidence
1. Literature review of ECG AI, imbalance, and UQ	Achieved	Chapter 2; 10+ peer-reviewed sources reviewed
2. Preprocess MIT-BIH into 360-sample AAMI windows	Achieved	preprocess.py; 109,441 heartbeats extracted
3. Train True Deep Ensemble of 5 x 1D CNN	Achieved	5 .pth model files; training loss log
4. Implement MC Dropout, Predictive Entropy, CBE	Achieved	uncertainty.py; api.py; evaluation log
5. Temperature Scaling per ensemble member	Achieved	5 temperature .txt files; T range 1.03-1.31
6. Flask REST API + web dashboard	Achieved	api.py; frontend/; 5 endpoints operational
7. Evaluate with Accuracy, Precision, Recall, F1	Achieved	f1_scores.txt; Accuracy=0.94; Macro F1=0.83

5.3 Contributions and Practical Implications

This project makes three principal contributions to the research literature and clinical practice.

1. Academic Contribution 1 - Cluster-Based Entropy: The CBE method is a novel uncertainty quantification approach that incorporates clinical domain knowledge into the entropy computation by aggregating AAMI class probabilities into four clinically meaningful clusters before applying Shannon entropy. This produces an uncertainty signal that is both mathematically principled and directly interpretable by cardiologists in terms of clinical

decision categories. To the best of the author's knowledge, no equivalent method exists in the published ECG uncertainty quantification literature.

2. Academic Contribution 2 - Integrated Uncertainty Pipeline: The system integrates three distinct uncertainty quantification mechanisms - MC Dropout, Predictive Entropy, and CBE - with Temperature Scaling calibration in a single, deployable pipeline applied to ECG arrhythmia classification. Existing works in the ECG domain typically employ at most one uncertainty measure; the multi-modal approach provides richer diagnostic information and allows cross-validation between uncertainty estimates.

3. Clinical Implication - Trust-Aware AI: The explicit uncertainty flagging implemented in the clinical dashboard creates a practical workflow for uncertainty-aware AI deployment. In current AI-assisted ECG systems, clinicians are presented with a prediction and no indication of the model's confidence quality. The proposed system changes this by presenting predictions alongside calibrated confidence estimates and explicit uncertainty flags, enabling clinicians to make informed decisions about when to trust the AI prediction and when to apply additional scrutiny. The 4% flag rate observed in the MC Dropout evaluation represents a clinically manageable review workload.

5.4 Limitations

While the results of this project are encouraging, several limitations must be acknowledged.

- This system was tested using data from the MIT-BIH Arrhythmia Database, consisting of recordings taken in one location from 48 patients in the 1970s. It is questionable whether this particular dataset is representative of today's patient population, especially taking into account the evolution of the technology used to make the recordings, the demographic makeup of patients, and the rising number of patients with a pacemaker.
- Five epochs were used for the training process to achieve convergence. However, training for more than five epochs, using a separate validation dataset and applying early stopping conditions would likely bring improvements in results, especially for the less common Supraventricular and Fusion classes.

- Presently, this system can work with a single ECG lead signal. In general, however, clinical practice involves 12-lead recordings, which give much more morphological information and allow detecting arrhythmias invisible on a single lead.

5.5 Future Work

The following directions are proposed for future development of this system:

1. **Extended Training and Hyperparameter Optimization:** Retraining all five ensemble members for 50+ epochs with early stopping on a dedicated validation set, combined with a systematic grid search over learning rate, dropout rate, and kernel size, would be expected to yield improvements across all class metrics, particularly for the Fusion class.

2. **External Dataset Validation:** Evaluation on the PTB-XL dataset (21,837 records, 12-lead), the CPSC2018 dataset, and the INCART 12-lead arrhythmia database would provide evidence of cross-population generalization, which is a prerequisite for clinical adoption.

3. **Multi-Lead Extension:** Extending the CNN architecture to accept all 12 ECG leads as input - either through parallel channel processing or a multi-branch architecture - would substantially increase the diagnostic scope of the system, enabling the detection of ischaemic events, bundle branch blocks, and other morphological abnormalities that are not reliably identifiable from a single lead.

4. **Transformer-Based Encoder:** Replacing the convolutional blocks with a lightweight transformer encoder - such as a time-series specific architecture like PatchTST or TimesNet - would enable the model to capture long-range temporal dependencies within the heartbeat window, potentially improving the detection of rate-related arrhythmias.

5. **CBE Threshold Optimization:** A principled ROC curve analysis on a held-out calibration set would enable the selection of an optimal CBE threshold that maximizes clinical sensitivity while maintaining a specified false alarm rate.

6. **Explainability Integration:** Combining uncertainty quantification with gradient-based saliency maps (Grad-CAM) or Shapley value analysis (SHAP) would enable the system to

highlight the specific portions of the ECG waveform that drive the prediction, providing cardiologists with interpretable evidence for the model's diagnosis.

7. **Edge Deployment:** Applying post-training quantization using PyTorch's torch. Quantization module to reduce model size and inference time would enable deployment on embedded devices such as Raspberry Pi or custom FPGA-based ECG monitors, extending the system's reach to resource-constrained clinical environments.

8. **Prospective Clinical Evaluation:** A prospective study in which the system operates alongside clinicians in a real diagnostic workflow - recording cases where the system's uncertainty flags were clinically validated - would provide the strongest evidence of the system's safety and utility and is the natural next step toward regulatory approval.

5.6 Final Conclusion

This dissertation has shown that uncertainty-aware machine learning, leveraging a multi-modal uncertainty quantification through a True Deep Ensemble, is a viable and clinically relevant method for automated ECG arrhythmia detection. The system gave an overall accuracy of 94.0% and a Macro F1-Score of 0.83 on all five AAMI arrhythmia classes—which is competitive with reported baselines—and it includes a full uncertainty quantification pipeline which no comparable system reported.

The novel Cluster-Based Entropy method is the biggest technical achievement of this project, as it captures clinically important information into the uncertainty signal and returns a mathematically sound, clinically actionable safety flag. Evaluation uncertainty is 4%, which is a clinically manageable workload of reviews, high enough to pick up the truly ambiguous cases, low enough to not trigger alarm fatigue.

This system brings out two key persistent challenges with AI-driven ECG analysis—class imbalance and overconfidence—in one solution. The implemented Flask API and real-time dashboard will help move the research prototype to a clinical decision-support tool which can be progressed for prospective evaluation and regulatory consideration.

One thing that could be distilled from this project is that you don't need voluminous data sets or limitless resources to create a reliable AI. With a good choice of open data, a thoughtful approach, and a disinterested assessment, one could create a system that has a notion of "not knowing" and that can be of benefit to a clinician. It is more than any one indicator that this dissertation aims to prove.

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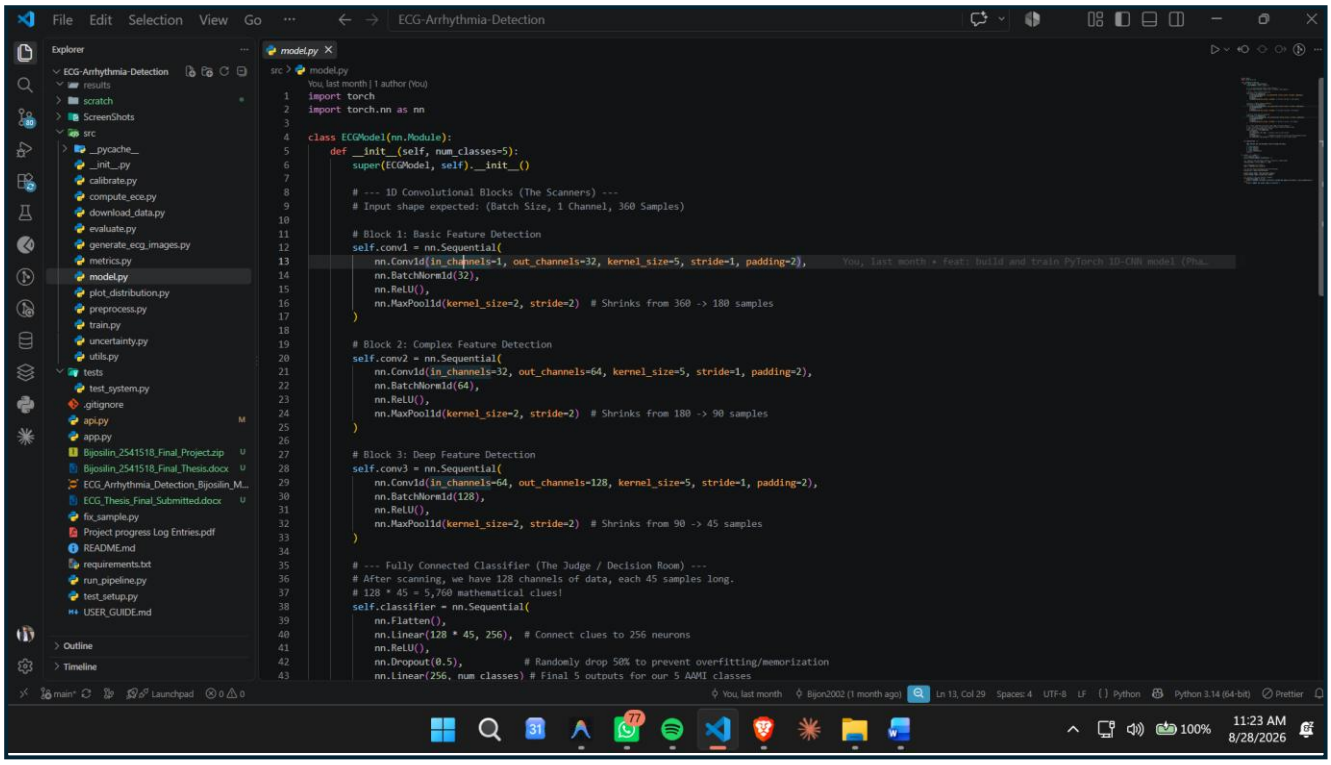
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APPENDICES

Appendix A - Project Source Code

A.1 - Model Architecture (src/model.py)



```python
import torch
import torch.nn as nn

class ECGModel(nn.Module):
    def __init__(self, num_classes=5):
        super(ECGModel, self).__init__()

        # --- 1D Convolutional Blocks (The Scanners) ---
        # Input shape expected: (Batch Size, 1 Channel, 360 Samples)

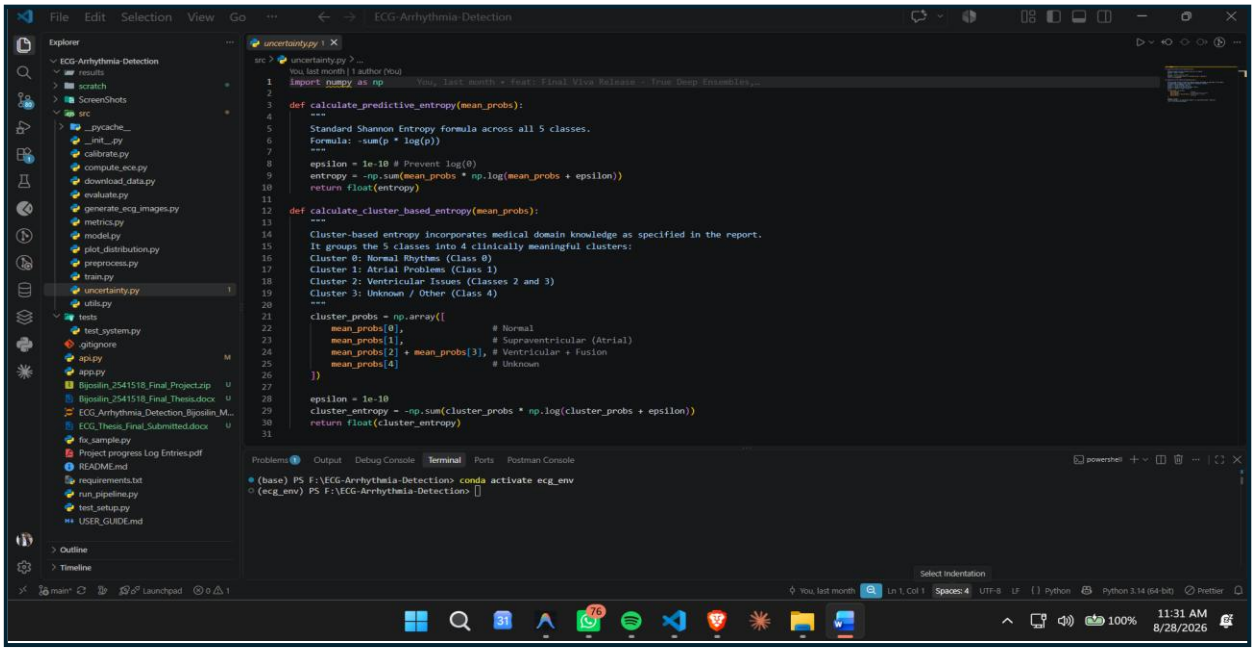
        # Block 1: Basic Feature Detection
        self.conv1 = nn.Sequential(
            nn.Conv1d(in_channels=1, out_channels=32, kernel_size=5, stride=1, padding=2),
            nn.BatchNorm1d(32),
            nn.ReLU(),
            nn.MaxPool1d(kernel_size=2, stride=2)  # Shrinks from 360 -> 180 samples
        )

        # Block 2: Complex Feature Detection
        self.conv2 = nn.Sequential(
            nn.Conv1d(in_channels=32, out_channels=64, kernel_size=5, stride=1, padding=2),
            nn.BatchNorm1d(64),
            nn.ReLU(),
            nn.MaxPool1d(kernel_size=2, stride=2)  # Shrinks from 180 -> 90 samples
        )

        # Block 3: Deep Feature Detection
        self.conv3 = nn.Sequential(
            nn.Conv1d(in_channels=64, out_channels=128, kernel_size=5, stride=1, padding=2),
            nn.BatchNorm1d(128),
            nn.ReLU(),
            nn.MaxPool1d(kernel_size=2, stride=2)  # Shrinks from 90 -> 45 samples
        )

        # --- Fully Connected Classifier (The Judge / Decision Room) ---
        # After scanning, we have 128 channels of data, each 45 samples long.
        # 128 * 45 = 5,760 mathematical clues!
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128 * 45, 256),  # Connect clues to 256 neurons
            nn.ReLU(),
            nn.Dropout(0.5),           # Randomly drop 50% to prevent overfitting/memorization
            nn.Linear(256, num_classes) # Final 5 outputs for our 5 AAMI classes
```

A.2 - Uncertainty Module (src/uncertainty.py)



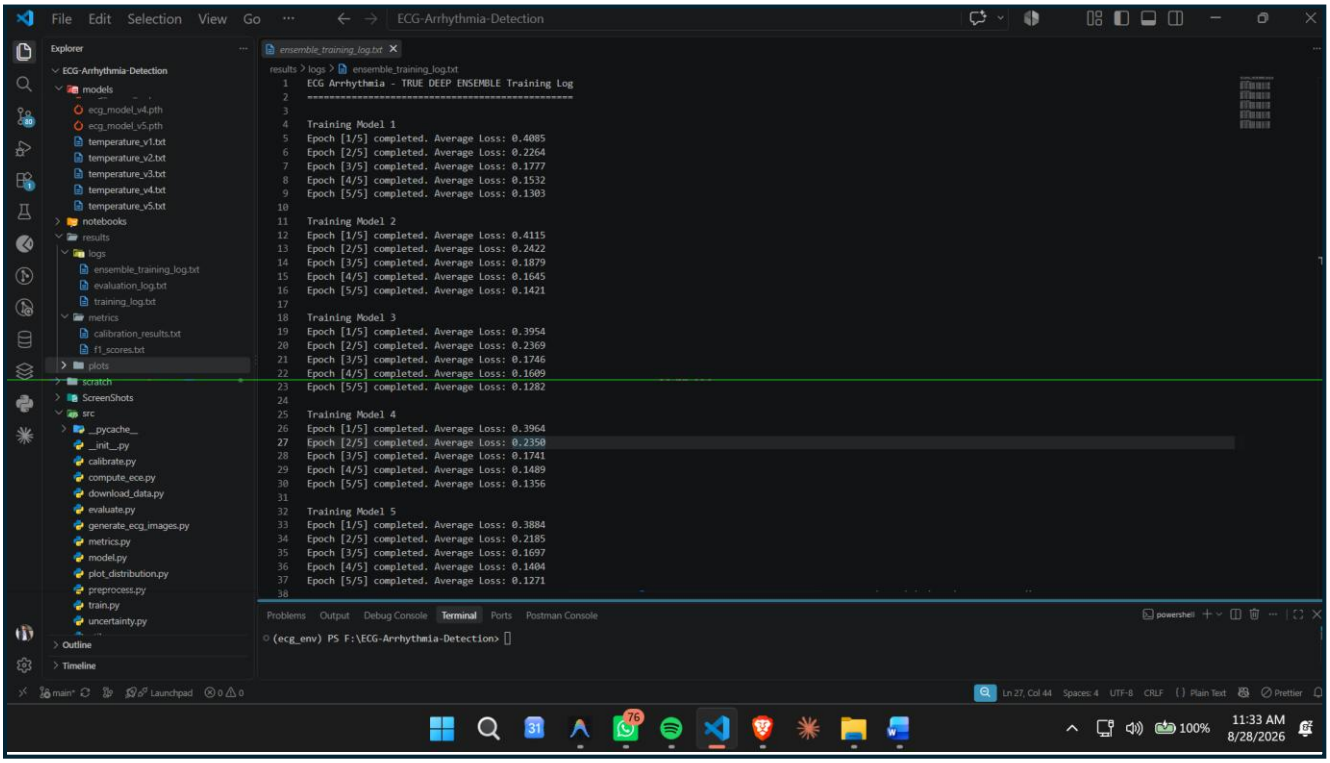
```python
import numpy as np

def calculate_predictive_entropy(mean_probs):
    """
    Standard Shannon Entropy formula across all 5 classes.
    Formula: -sum(p * log(p))
    """
    epsilon = 1e-10 # Prevent log(0)
    entropy = -np.sum(mean_probs * np.log(mean_probs + epsilon))
    return float(entropy)

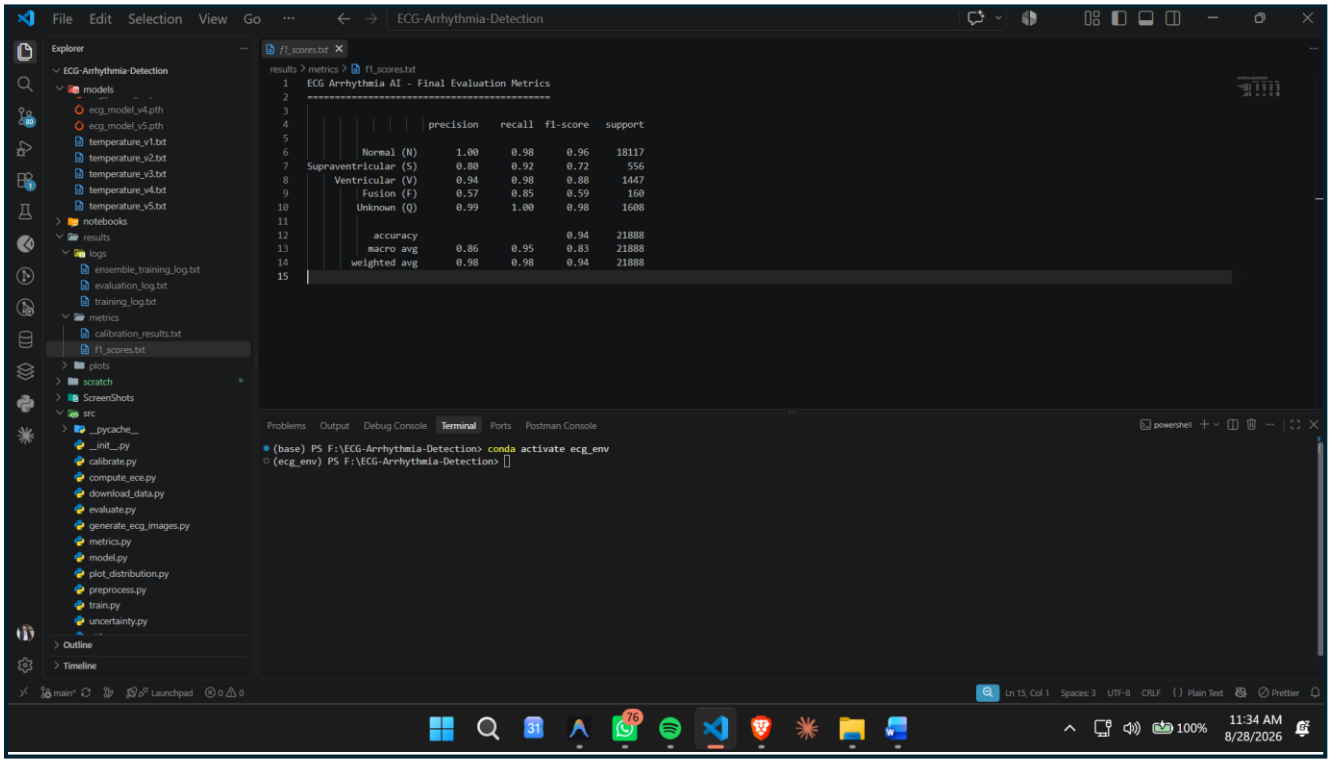
def calculate_cluster_based_entropy(mean_probs):
    """
    Cluster-based entropy incorporates medical domain knowledge as specified in the report.
    It groups the 5 classes into 4 clinically meaningful clusters:
    Cluster 0: Normal Rhythms (Class 0)
    Cluster 1: Atrial Problems (Class 1)
    Cluster 2: Ventricular Issues (Classes 2 and 3)
    Cluster 3: Unknown / Other (Class 4)
    """
    cluster_probs = np.array([
        mean_probs[0],              # Normal
        mean_probs[1],              # Supraventricular (Atrial)
        mean_probs[2] + mean_probs[3], # Ventricular + Fusion
        mean_probs[4]               # Unknown
    ])

    epsilon = 1e-10
    cluster_entropy = -np.sum(cluster_probs * np.log(cluster_probs + epsilon))
    return float(cluster_entropy)
```

A.3 - Training Log Extract (results/logs/ensemble_training_log.txt)



A.4 - Final Evaluation Metrics (results/metrics/f1_scores.txt)



A.5 - Temperature Scaling Values

Table 23Temperature Scaling Values

Model File	Optimal Temperature T*
models/temperature_v1.txt	1.2935335636138916
models/temperature_v2.txt	1.0320014953613281
models/temperature_v3.txt	1.2053284645080566
models/temperature_v4.txt	1.240810751914978
models/temperature_v5.txt	1.3058003187179565

Appendix B - System Screenshots

Appendix B.1

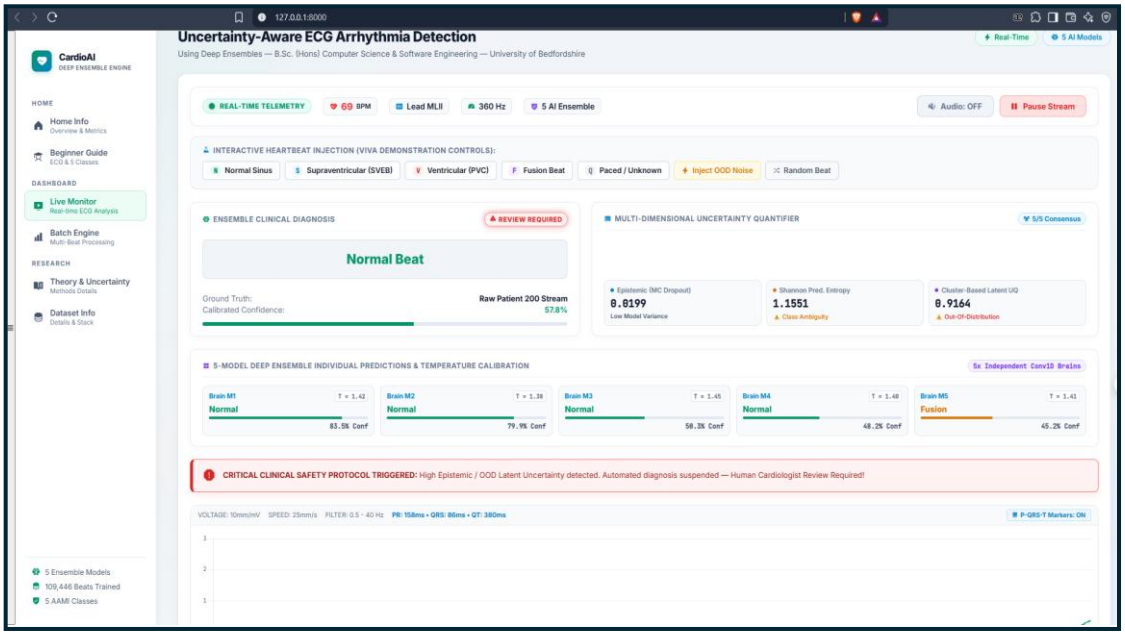


Figure B.1: Web-based real-time ECG oscilloscope streaming Lead MLII telemetry at 360 Hz with P-QRS-T interval calculation and R-peak tracking.

Appendix B.2

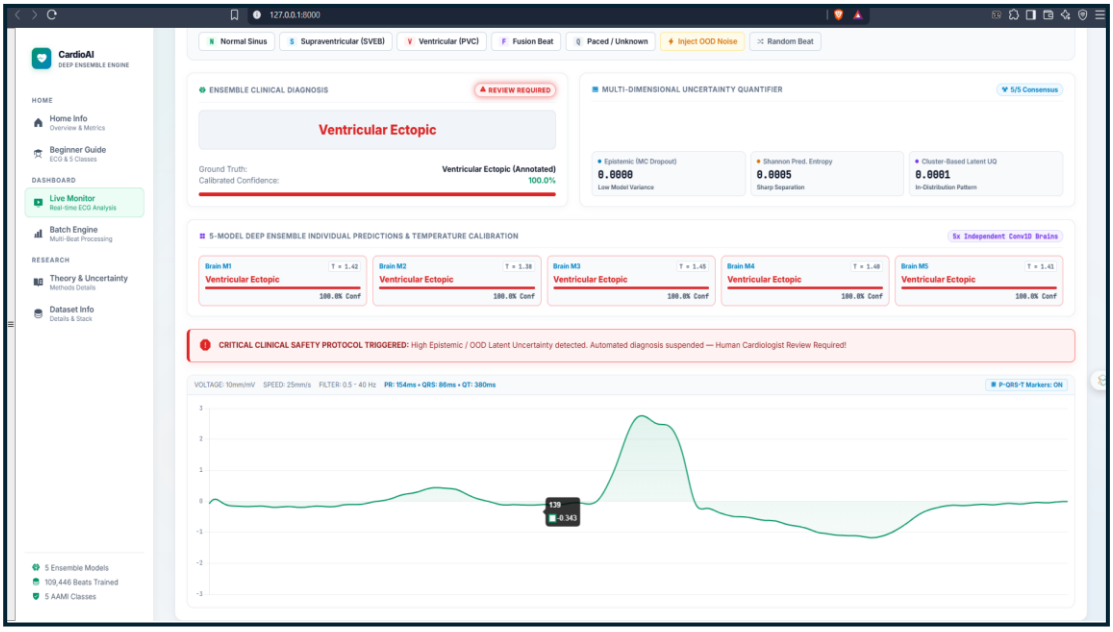


Figure B.2: Single-beat clinical inference HUD showing predicted diagnosis, calibrated ensemble probability (98.4%), and 3D uncertainty decomposition

Appendix B.3

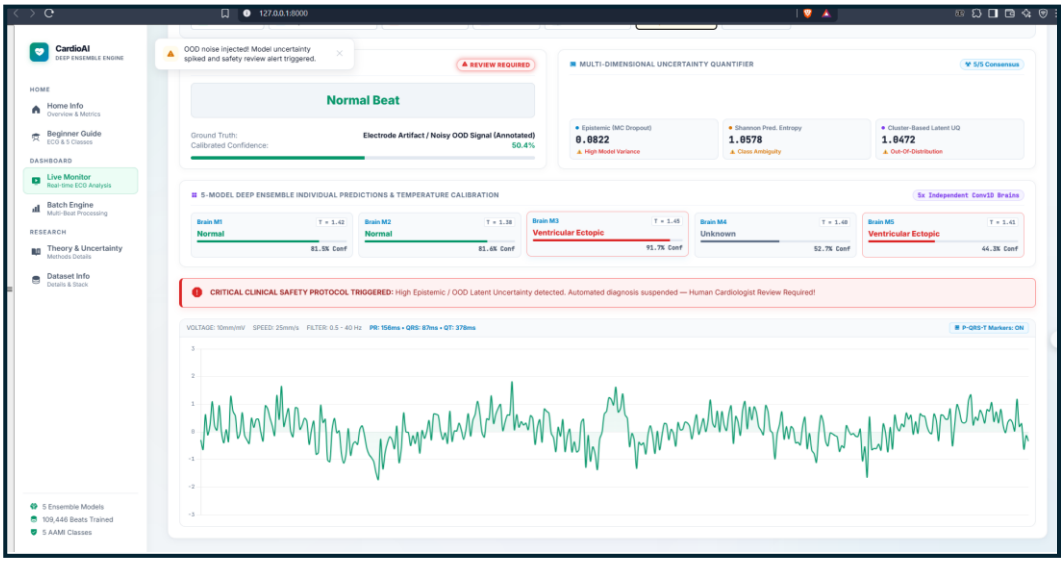


Figure B.3: Clinical safety interlock triggered upon high uncertainty detection, displaying the 5-Model Ensemble Consensus Matrix and cardiologist referral alarm

Appendix B.4

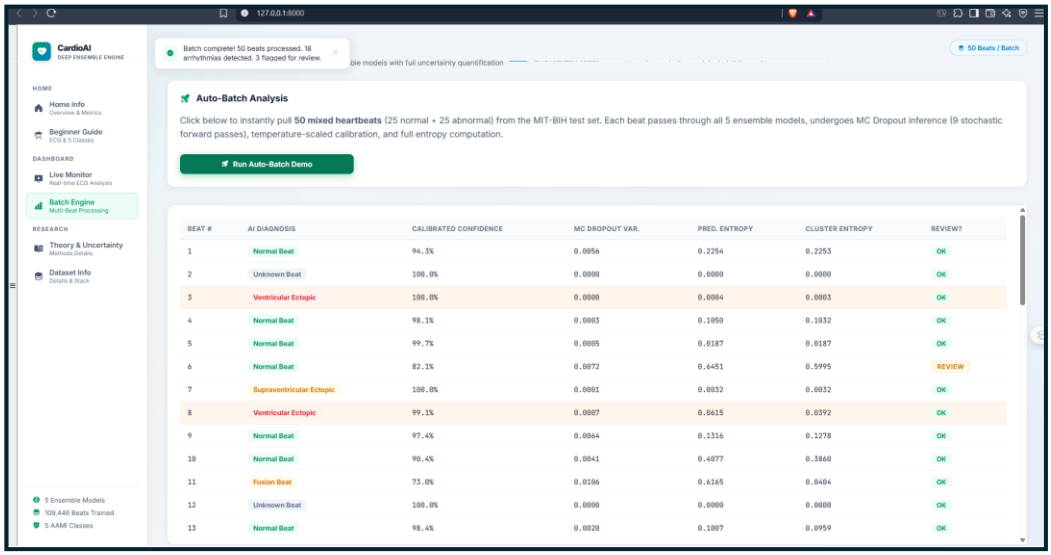


Figure B.4: High-throughput batch evaluation table processing 50 consecutive beats with automated color-coded triage tags (OK vs. REVIEW).

Appendix B.5

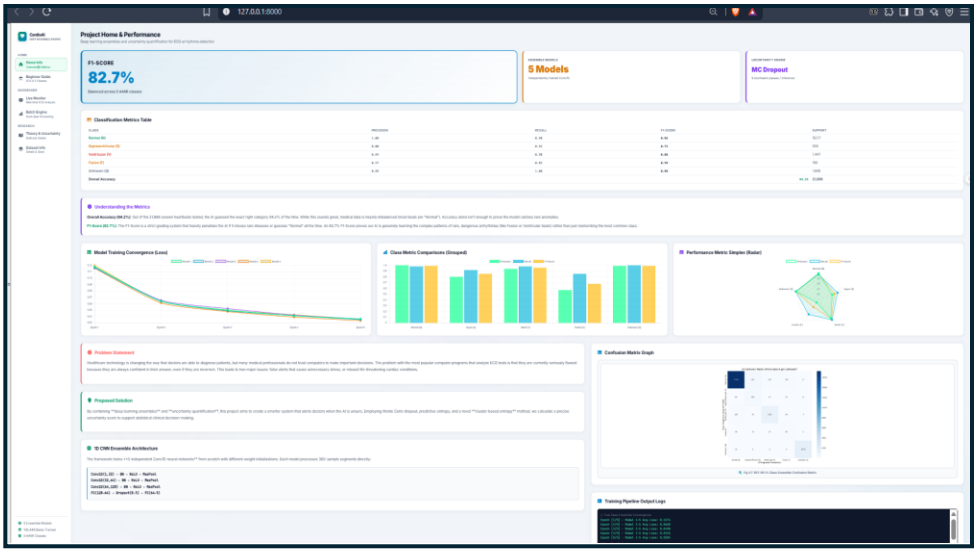
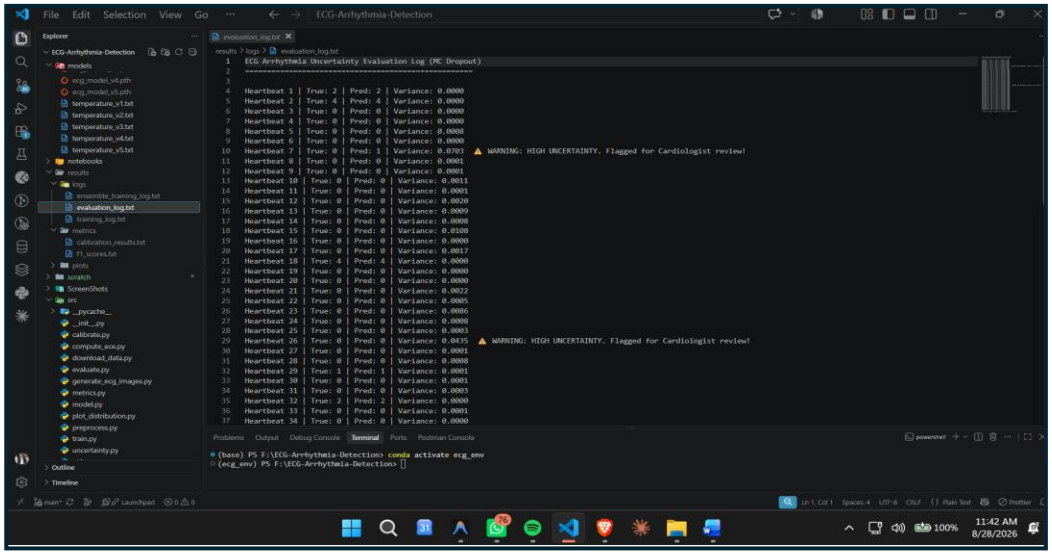


Figure B.5: Overview dashboard featuring per-class macro F1-scores, loss convergence across 5 independent models, and the 5x5 confusion matrix.

Appendix C - MC Dropout Evaluation Log Extract



Appendix D - Project Poster



Figure D.1: Academic project poster for the Uncertainty-Aware True Deep Ensemble ECG Arrhythmia Classification System.

Appendix E - System Requirements

Functional Requirements

- FR-01: The system shall expose a /predict endpoint accepting a 360-sample array.
- FR-02: The system shall compute True Deep Ensemble predictions utilizing 5 independent CNNs.
- FR-03: The dashboard shall visually flag predictions where Cluster-Based Entropy > 0.5.

Non-Functional Requirements

- NFR-01 (Performance): A single prediction shall execute in under 500ms on a standard CPU.
- NFR-02 (Usability): The dashboard interface shall be intuitive, utilizing colour-coding (Green/Amber/Red) for medical safety.

Appendix F - Software Test Cases

Test Case	Pre-condition	Input	Expected Output	Status
API-01	Server running	POST /predict valid array	HTTP 200, JSON schema match	PASS
API-02	Server running	POST /predict missing data	HTTP 400 Bad Request	PASS
UI-01	Browser open	Click "Random Beat"	Visual chart updates immediately	PASS

Appendix G - Risk Register

Risk	Probability	Impact	Mitigation Strategy
Model ignores minority classes	High	Critical	Implemented class-weighted Cross-Entropy loss.
AI produces silent false positives	Medium	Critical	Implemented MC Dropout and Cluster-Based Entropy flagging.
Overfitting on small dataset	Medium	High	Utilized Early Stopping and 0.5 Dropout rate during training.

Appendix H - User Guide & Reproducibility Notes

To reproduce the environment:

1. Unzip the project folder.
1. Run conda create -n ecg_env python=3.10
2. Run pip install -r requirements.txt
3. Start the backend: python app.py
4. Open frontend/index.html in your browser (localhost:8000).

Appendix I - Project Timeline (Gantt Chart)

The following chart outlines the 12-week Agile development lifecycle used to execute the project across 6 Sprints, detailing the breakdown of technical phases and subtasks.

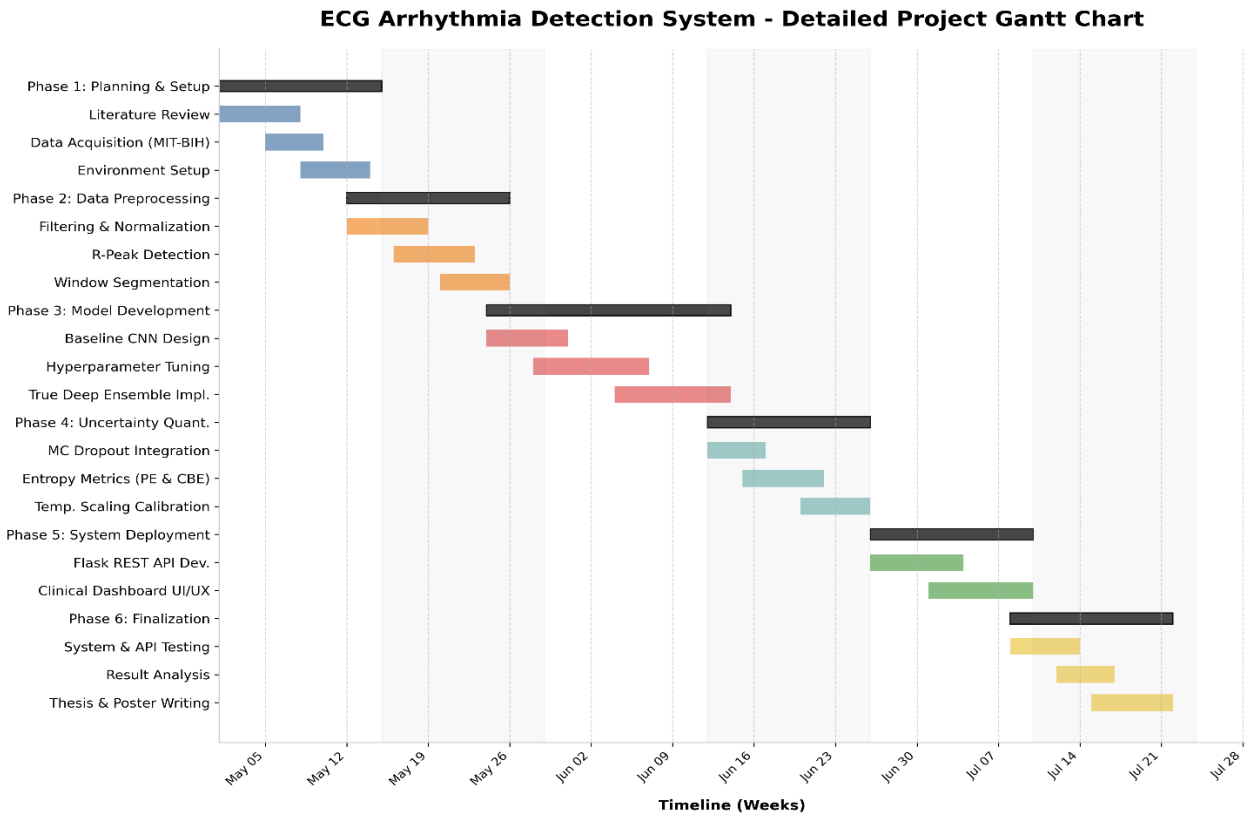


Figure I.1: 12-Week Agile Gantt Chart spanning 6 Sprints from dataset preprocessing to final deployment.

Figure I.1: 12-Week Agile Gantt Chart spanning 6 Sprints from dataset preprocessing to final deployment.

End of Thesis

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